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NOMAD CORTEX Key Management Operations Guide

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Revision History

Version	Date	Reason	Who
1.0	22/04/98	Initial Accell version	P Kingston
2.0	24/03/04	First full J2EE, 3DES version	P Kingston
2.1	07/06/04	TMK replacement enhancements	P Kingston
2.2	08/06/04	Further TMK changes	P Kingston
2.3	02/09/04	Incorporated dynamic links to User Guide source material where appropriate. Added description for encrypting clear ZMK components under the LMK.	P Kingston
2.4	06/04/05	Added CSCKs Added Maintaining EMV Cryptogram Version Definitions	P Kingston
2.5	17/09/05	Clarified how to specify two-part TMK printing. Integrated parts of the separate *EMV Key Management* document.	P Kingston

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Introduction

Purpose

NOMAD CORTEX ensures the integrity and security of transactions, and of communications between various parties using *encryption keys*. NOMAD CORTEX supports a standard key hierarchy and key management procedures.

Encryption is central to the performance of the whole NOMAD CORTEX network. The purpose of this document is to ensure that encryption keys are implemented effectively.

The procedures described in this document can also be found in the appropriate User Guide.

Functions

NOMAD CORTEX supports the following encryption activities:

- Generating and storing keys encrypted under an LMK
- Viewing the status of encryption keys
- Generating and storing ZMK components
- Generating and storing ZPKs encrypted under a ZMK
- Generating and storing TMKs
- Generating TPKs
- Generating and storing CVKs and PVKs
- Generating and storing IMKs
- Generating and storing MAC keys (TAKs)

Scope

- NOMAD CORTEX supports both *Single DES* and *Triple DES (3DES)* encryption functionality where appropriate (provided by an *HSM*). This document, however, is primarily concerned with 3DES operations.
- This document describes both Europay (now *MasterCard Europe*) and Visa encryption functionality. Both networks deal with encryption keys in a similar way—there are differences, however, in the terminology used. Where such differences exist, these are highlighted. NOMAD CORTEX satisfies the requirements of Visa International and of MasterCard Europe.
- This document describes only the most commonly performed HSM procedures. If you can’t find a description for an HSM procedure, please refer to the Thales HSM documentation.

Note: The following sections assume that you are familiar with the basics of using the NOMAD CORTEX system, and with the NOMAD Software documentation conventions. If you are not, please refer to the document, *Getting Started with NOMAD CORTEX*.

Abbreviations

The following abbreviations are used in this document:

Term	Description
3DES	Triple-DES (Data Encryption Standard)
ACMK	Application Cryptogram Master Key
ARPC	Authorisation Response Cryptogram
ARQC	Authorisation Request Cryptogram
AWK	Acquirer Working Key
CAVV	Cardholder Authentication Verification Value
CVC	Card Verification Code
CVK	Card Verification Key
CVV	Card Verification Value
CVV2	Card Verification Value 2
DAC	Data Authentication Code
DACMK	Data Authentication Code Master Key
DDA	Dynamic Data Authentication
DES	Data Encryption Standard
DN	Dynamic Number
DNMK	Dynamic Number Master Key
EMV	Europay, MasterCard and Visa
IWK	Issuer Working Key
LMK	Local Master Key
LZPK	Local Zone PIN Key
MAC	Message Authentication Code
PVK	PIN Verification Key
PVV	PIN Verification Value
SDA	Static Data Authentication
SMCMK	Secure Messaging with Confidentiality Master Key
SMIMK	Secure Messaging with Integrity Master Key
TMK	Terminal Master Key
TPK	Terminal PIN Key
ZMK	Zone Master Key
ZPK	Zone PIN Key

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1. Overview of Encryption Concepts

Essentially, *encryption* is the conversion of data into another format that cannot be easily understood by those not authorised to view it. *Decryption* is therefore the process of converting this encrypted data back into its original form, so it can be understood.

Simple encryption methods include the substitution of letters for numbers and the exchanging of letters in the alphabet. In secure commercial applications, more complex encryption methods work according to sophisticated algorithms that rearrange the data.

To easily recover the content of a piece of encrypted data, the correct decryption *key* is required. This key is essentially an algorithm that reverses the work of the encryption algorithm. Recently, computers have been used to 'break' the code. The more complex the encryption algorithm, the more difficult it becomes to eavesdrop on the communications without access to the key.

Encryption/decryption is essential when carrying out any kind of sensitive transaction, such as a credit card purchase online. Whilst there are many kinds of encryption employed in electronic commerce, they can basically be split into two methodologies, *Symmetric* and *Asymmetric* encryption.

1.1. Symmetric and asymmetric encryption

1.1.1. Symmetric encryption

In symmetric encryption, the message can be encrypted and decrypted using the same key. Compared to asymmetric encryption, (which is described later), symmetric encryption is fast. However, a disadvantage in symmetric encryption is that both the sender and the recipient must have an access to same encryption key. In other words, secure distribution of the encryption key between the parties is required. The key can be distributed personally, for instance, on a disk, or it can be delivered encrypted, or in some other secure way.

A commonly used symmetric encryption method is *Triple DES (3DES)*, which is developed from *DES (Data Encryption Standard)* (see below).

1.1.1.1. DES

The *Data Encryption Standard (DES)* was developed by IBM during the 1970s. Unfortunately, over time various ways were found that could significantly reduce the amount of time needed to find a DES key by 'brute force' (i.e., by using massive computing power). In 1998, the Electronic Frontier Foundation, using a specially developed computer called the DES Cracker, managed to break DES in less than 3 days. And this was done for under $250,000!

As computers become progressively faster and more powerful, it has been recognised that a 56-bit key is simply not large enough for high-security applications. However, despite the growing concerns about its vulnerability, DES is still widely used by financial services and other industries worldwide to protect sensitive online applications.

DES encrypts and decrypts data in 64-bit blocks. It takes a 64-bit block of plain data as input and outputs a 64-bit block of encrypted data.

1.1.1.2. 3DES

3DES answers many of the shortcomings of DES. As it is based on the DES algorithm, it is relatively easy to modify existing software to use 3DES. It also has the advantage of proven reliability and a longer key length that eliminates many of the shortcut attacks that can be used to reduce the amount of time it takes to break DES.

3DES uses keys that are twice as long, and that effectively means it will take an average of 2^{56} times longer to crack. The procedure for encryption is exactly the same as regular DES, but it is repeated three times—hence the name ‘Triple’ DES.

If we consider a triple-length key to consist of three keys, Key 1, Key 2, and Key 3 then encryption using 3DES is as follows:

- Encrypt with Key 1
- Decrypt with Key 2
- Encrypt with Key 3

Decryption is the reverse process. However, like most card processing systems—including Visa and MasterCard—NOMAD CORTEX does not make use of triple-length keys, only double and single. For double-length keys, therefore, Key 3 is set to equal Key 1, and for single length, Keys 1, 2 and 3 are all equal.

Due to the extra processing involved, 3DES runs three times slower than standard DES, but is much more secure if used properly. Both Visa and MasterCard have mandated use of 3DES, and NOMAD CORTEX is ready for this.

1.1.2. Asymmetric encryption

Asymmetric encryption is based on the use of key pairs; where one part of a key pair is public and another part is private (hence asymmetric encryption is also called *public key encryption*). The keys are exchangeable, which means that a message encrypted with the public key can be decrypted using the private key and vice versa. For instance, an e-mail message can be encrypted with the recipient's public key and the recipient decrypts the message with the private key. Since the recipient's private key is only in the recipient's possession, no third party is able to decrypt the message encrypted with the recipient's public key.

Asymmetric encryption's advantage is the management of keys, as private keys need not to be distributed to the recipient. On the other hand, asymmetric encryption is time-consuming.

NOMAD CORTEX does not currently make use of asymmetric encryption, although the RSA public/private keys used for communication between smart cards and EMV-capable terminals do.

1.2. Encryption keys

Encryption keys are basically variable values that are applied using an algorithm to a string or block of unencrypted data to produce encrypted data, or to decrypt encrypted data. There are many different encryption keys used by NOMAD CORTEX, but broadly speaking, they can be grouped into two distinct types:

- PIN keys
- Master keys

There are keys (such as the *MAC Key*) that do not belong to either of these groups. This is discussed in more depth later.

PIN keys are used to encrypt the PINs that are used for cardholder authentication.

Master keys are used to encrypt PIN keys when they need to be exchanged between different entities. Therefore PIN keys cannot be generated or communicated without Master keys being established.

PIN keys and Master keys can exist in different *Zones*:

- *External Zone*—Between parties (Institutions) and across separate (shared) networks
- *Internal Zone*—Within the NOMAD CORTEX system
- *Terminal Zone*—Between NOMAD CORTEX and a terminal, and across an internal managed network

The location of these Zones is shown in the figure below:

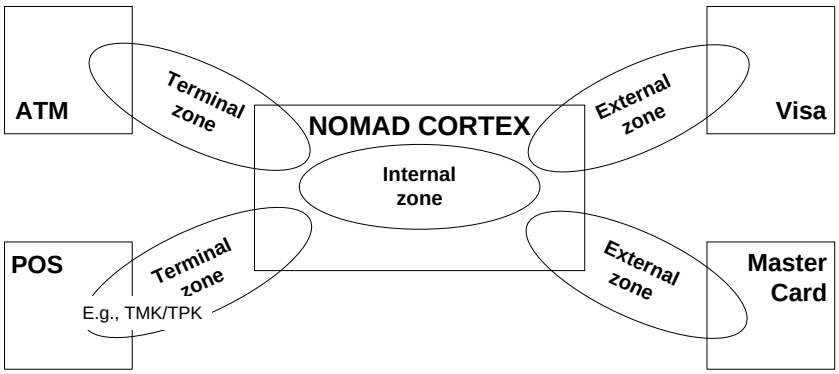


Figure 1: NOMAD CORTEX Encryption Zones

Referring to the figure, the *Terminal Master Key* (*TMK*) is an example of a Master key in the Terminal Zone. An example of a PIN key in the Terminal Zone is the *Terminal PIN Key* (*TPK*). The TPK is used to encrypt the PIN block for transport to NOMAD CORTEX, and is itself encrypted under the TMK for transport from NOMAD CORTEX to the terminal.

Note: Visa and MasterCard deal with encryption keys in a similar way—there are differences, however, in the terminology used. Where such differences exist, these are highlighted. NOMAD CORTEX satisfies the requirements of Visa International and of MasterCard Europe.

Within NOMAD CORTEX, the following other encryption keys are used to ensure data security and integrity:

1.2.1. Local Master Keys

The *Local Master Keys (LMKs)* are used during system installation to initialise the entire encryption sub-system. LMKs are always used in pairs. All other stored keys are encrypted under an LMK pair, and LMKs must exist before any other option can be used. This is shown in the figure below:

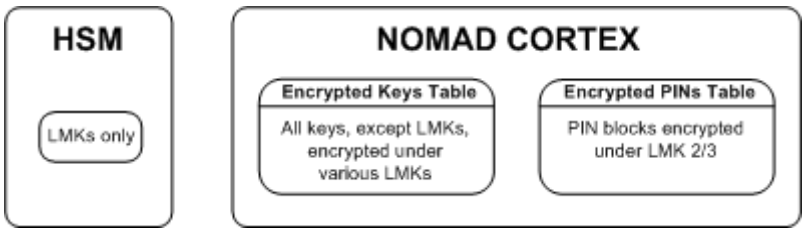


Figure 2: LMKs within NOMAD CORTEX and the HSM

To generate a set of LMKs, typically, three security officers are appointed, with each inventing two sequences of 16 characters consisting of characters in the ranges 0-9 and A-F (the LMK *components*). These sequences are then stored both on the Host Security Module (HSM) and on three password-protected ‘smart’ cards. At no time does any one person have access to all LMK components. The precise method used for generating LMKs, however, will vary between HSM manufacturers, and for the procedure to follow for your HSM you should refer to the appropriate manufacturer documentation.

LMKs are never exchanged between other parties.

1.2.2. Zone Master Key (*Visa: Zone Control Master Key*)

Note: Visa's terminology for a Zone Master Key is *Zone Control Master Key (ZCMK)*. Any reference in this document to Zone Master Keys also applies to ZCMKs, except where indicated otherwise.

The *Zone Master Key (ZMK)* is used for sharing *Zone PIN Keys* (*Visa: Issuer Working Key/Acquirer Working Key*) securely. The ZMK is usually exchanged with a network if PINs are to be passed between NOMAD CORTEX and that network. The ZMK is actually received as three clear (i.e., unencrypted) components, typically delivered separately by hand or by post, which are then encrypted under an LMK pair before being combined within the HSM to generate a single ZMK. Typically, three high-ranking bank officers will each enter a component. The encrypted components are then stored in NOMAD CORTEX.

The ZMK is stored in NOMAD CORTEX under the LMK (pair 04-05).

1.2.3. Terminal Master Key

The *Terminal Master Key (TMK)* is used for distributing other encrypting keys within a local network to terminals (including the *Terminal PIN Key* and *MAC Key*). In line with Visa mandates, NOMAD CORTEX requires a TMK to be set up for each terminal, i.e., a TMK will need to be set up each time a new terminal is added to the system.

The TMK is stored in NOMAD CORTEX under the LMK (pair 14-15).

1.2.4. Zone PIN Key (*Visa: Issuer/Acquirer Working Key*)

Note: Visa's terminology for a *Zone PIN Key* is *Issuer Working Key (IWK)*, for incoming messages and *Acquirer Working Key (AWK)*, for outgoing messages. Any reference in this document to Zone PIN Keys also applies to IWK/AWKs, except where indicated otherwise.

The *Zone PIN Key (ZPK)* is used to encrypt PINs for transfer between communicating parties, i.e., between acquirers and issuers. The ZPK is distributed automatically. With respect to EMV issuing, the ZPK is used to transfer the PIN block of a card between NOMAD CORTEX and the card personalisation system.

For exchange, a ZPK is encrypted under a ZMK; and for local storage, under an LMK (pair 06-07).

For exchange, IWKs and AWKs are encrypted under the Visa ZMK, and for local storage, under the LMK (pair 24-25).

1.2.5. Local Zone PIN Key

See *Zone PIN Key* above. The Local Zone PIN Key (LZPK) is used for encryption of PIN data within NOMAD CORTEX.

The LZPK is stored in NOMAD CORTEX under the LMK (pair 06-07).

1.2.6. Terminal PIN Key

The *Terminal PIN Key (TPK)* is used to encrypt the customer PIN for its transmission from the terminal to NOMAD CORTEX. This key is automatically generated by NOMAD CORTEX each time a TMK is created, and is automatically sent to the terminal at each boot session, or at any time the terminal is brought into the open state. TPK generation requires no User interaction. For exchange, a TPK is encrypted under a TMK; and for local storage, under the LMK (pair 14-15).

1.2.7. MAC Key (Thales: TAK)

Message Authentication Codes (MACs) are used to prevent critical parts of a message from being changed. An encryption key (the *MAC Key*) is used to encrypt sensitive data (such as the transaction amount). Typically, the data is split into a number of blocks. The first block is encrypted using the MAC Key. The result is then 'XOR 'ed' with the second block, with the result being encrypted again using the MAC key. This procedure continues until all data blocks have been encrypted. The results of all intermittent stages are then discarded and the result is the MAC.

Like a TPK, this key is automatically generated by NOMAD CORTEX each time a TMK is created, and is automatically sent to the terminal at each boot session, or at any time the terminal is brought into the open state. MAC Key generation requires no User interaction.

Note: Thales refers to MAC Keys as *Terminal Authentication Keys (TAKs)*.

For exchange, a MAC Key is encrypted under a TMK; and for local storage, under the LMK (pair 16-17).

1.2.8. Card Verification Key

The *Card Verification Key (CVK)* is used to encrypt data by producing *Card Verification Values (CVVs)* for each BIN. CVKs are always single-length keys—they do not use 3DES variants. They are created in pairs using NOMAD CORTEX, and are effectively two halves of a double-length key.

The CVK is exchanged with a network to enable it to perform stand-in verification.

For exchange, the CVK is encrypted under the ZMK, and for local storage, under the LMK (pair 14v-15v).

1.2.9. PIN Verification Key

The *PIN Verification Key (PVK)* is used to encrypt and verify PIN verification data for each BIN. Like CVKs, PVKs are always single-length keys—they do not use 3DES variants. They are created in pairs using NOMAD CORTEX, and are effectively two halves of a double-length key.

The PVK is exchanged with a network to enable it to perform stand-in verification.

For exchange, the PVK is encrypted under the ZMK, and for local storage, under the LMK (pair 14-15).

1.2.10. Card Security Code Key

The *Card Security Code Key (CSCK)* is specific to American Express. It is used to encrypt data by producing *Card Security Codes (CSCs)* for each BIN. This is comparable to the relationship between CVKs and CVVs. CSCKs are always double-length keys.

For exchange, CSCKs are encrypted under the ZMK, and for local storage, under the LMK (14).

1.2.11. EMV Issuer Master Keys

Issuer Master Keys (IMKs) are specific to *EMV*. Sometimes they are also referred to as Master derivation Keys (MKs). For a detailed description of how these keys are generated and managed, refer to the document *EMV2000 Integrated Circuit Card Specification for Payment Systems Book 2 - Security and Key Management, Version 4.0, December, 2000*.

By way of summary, Issuer Master Keys are used to derive keys that are loaded on the EMV chip of a card during personalisation. They are typically exchanged with card production systems or bureaux.

When entering the IMK, it is important that the *Key Sequence Number* matches the *Derivation Key Index* coded on the cards during personalisation. Which index to use is something that should be agreed with the personalisation centre prior to the generation of the cards. The use of the Derivation Key index means that different batches of cards issued with the same BIN can use totally different keys. This is particularly important if cards are to be produced by different personalisation centres.

For exchange, IMKs are encrypted under the ZMK, and for local storage, under the LMK (pair 28-29).

1.2.11.1. Application Cryptogram Master Key

The *Application Cryptogram Master Key (ACMK)* is used by ICC Application Cryptograms. Briefly, the ACMK is used as one component to derive the unique-per-card keys used by EMV cards to generate the ARQC in request messages, and to verify the ARPC values in messages.

1.2.11.2. Secure Messaging with Integrity Master Key

The *Secure Messaging with Integrity Master Key (SMIMK)* is an IMK for generating the MAC at the end of issuer scripts.

1.2.11.3. Secure Messaging with Confidentiality Master Key

The *Secure Messaging with Confidentiality Master Key (SMCMK)* is an IMK for encrypting data in issuer script commands (e.g., a creating a new PIN).

1.2.11.4. Data Authentication Code Master Key

The *Data Authentication Code Master Key (DACMK)* is an IMK for generating and validating the *Data Authentication Code (DAC).* A valid DAC proves that *Offline Static Data Authentication (SDA)* was correctly performed.

1.2.11.5. Dynamic Number Master Key

The *Dynamic Number Master Key (DNMK)* is an IMK for validating *Dynamic Numbers (DNs).* A valid DN proves that the terminal has correctly performed offline *Dynamic Data Authentication (DDA).*

1.2.12. Key hierarchy

The table below details the hierarchy of the encryption keys described in the previous sections:

Level	Key	Single/ Double	Generator	Exchange under	Store in CORTEX under
1	LMK	Double	Bank	N/A	HSM/smart cards
2	ZMK	Double	Network/Bank/other party	Clear, 3 parts	LMK 04-05
	TMK	Double	Bank	Clear, 2 parts	LMK 14-15
	IMKs	Double	Network/Bank/other party	ZMK	LMK 28-29
3	ZPK	Double	Bank	ZMK	LMK 06-07
	LZPK	Double	Bank	N/A	LMK 06-07
	TPK	Double	Bank	TMK	LMK 14-15
	MAC	Double	Bank	TMK	LMK 16-17
	AWK	Double	Visa	ZMK	LMK 24-25
	IWK	Double	Visa	ZMK	LMK 24-25
	PVK	Single	Network/Bank/other party	ZMK	LMK 14-15
	CVK	Single	Network/Bank/other party	ZMK	LMK 14v-15v

This hierarchy is further illustrated in Figure 3:

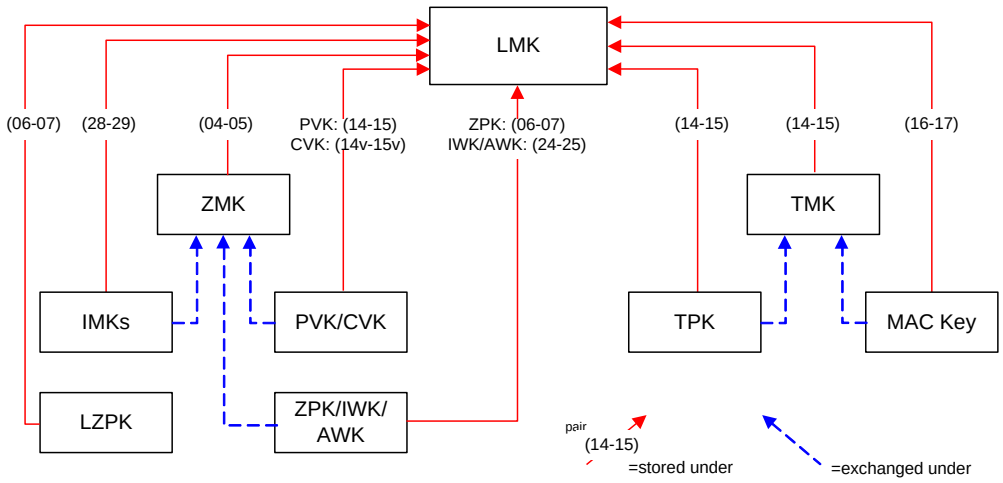


Figure 3: NOMAD CORTEX Encryption Key Hierarchy

2. Installing, Configuring and Initialising the HSM

NOMAD CORTEX interfaces to one or more Thales *Host Security Modules (HSMs)* to carry out encryption functions. The following sections describe how to take the HSM from its shipped state to online and functioning.

The HSM is a tamper-resistant device that provides the cryptographic facilities necessary for securing transactions in financial networks. Within a NOMAD CORTEX system, the HSM typically has the following uses:

- Generating encryption keys, such as LMKs, other Master keys (such as the TMK) and PIN keys (such as the TPK)
- Encrypting the PIN and Master keys generated under the LMK for storage
- Performing PIN translation, in other words, taking the PIN encrypted under the local ZPK and comparing it with that stored under the LMK
- Generating encrypted card values such as CVVs for payment cards
- Generating/verifying MACs for messages

Note: NOMAD CORTEX 2.6 was tested using the HSM8000 from Thales. It is also compatible with the HSM 7xxx series.

The flowchart below summarises the main steps involved in taking the HSM from its shipped state to online and functioning (relevant section numbers are shown in the bottom left of each box):

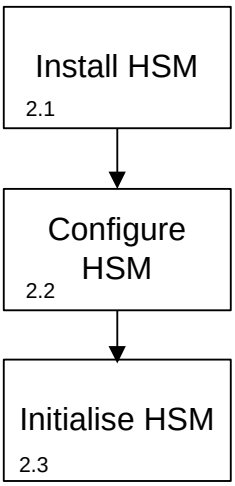


Figure 4: Taking the HSM from Shipped State to Online and Functioning

2.1. Installing the HSM

The following installation instructions relate to the Thales HSM8000.

Note: It is essential that you read *Appendix A - Warnings, Cautions and Statutory Statements* in the *Host Security Module 8000 Installation Manual* before proceeding.

Figure 5: HSM8000: Front View showing Normal Operation

An HSM system consists of a single stand-alone unit or a number of units mounted in a standard 19-inch cabinet. The HSM8000 can also be used to complement other RG7000 series HSMs in a standard 5-unit cabinet. A typical five-unit configuration permits concurrent operation for high throughput, and, under control of the application program, provides automatic and immediate backup in the event of a fault in a single unit.

The HSM is normally online to the Host and does not require operator monitoring or intervention. The HSM performs cryptographic processing in response to commands from the Host. The Host sends command messages, which consist of command codes and other fields that are required by the HSM in order to process the commands. The HSM processes the command messages and generates response messages, which also contain a variable number of fields (depending on the message type). Some commands, mainly involving plain text data, are entered by the User via the associated HSM Console.

The throughput of the HSM depends on the types of commands that are executed, and the method and speed of the Host connection.

Note: The console terminal, printer, cables and transceiver are **not** supplied with the HSM

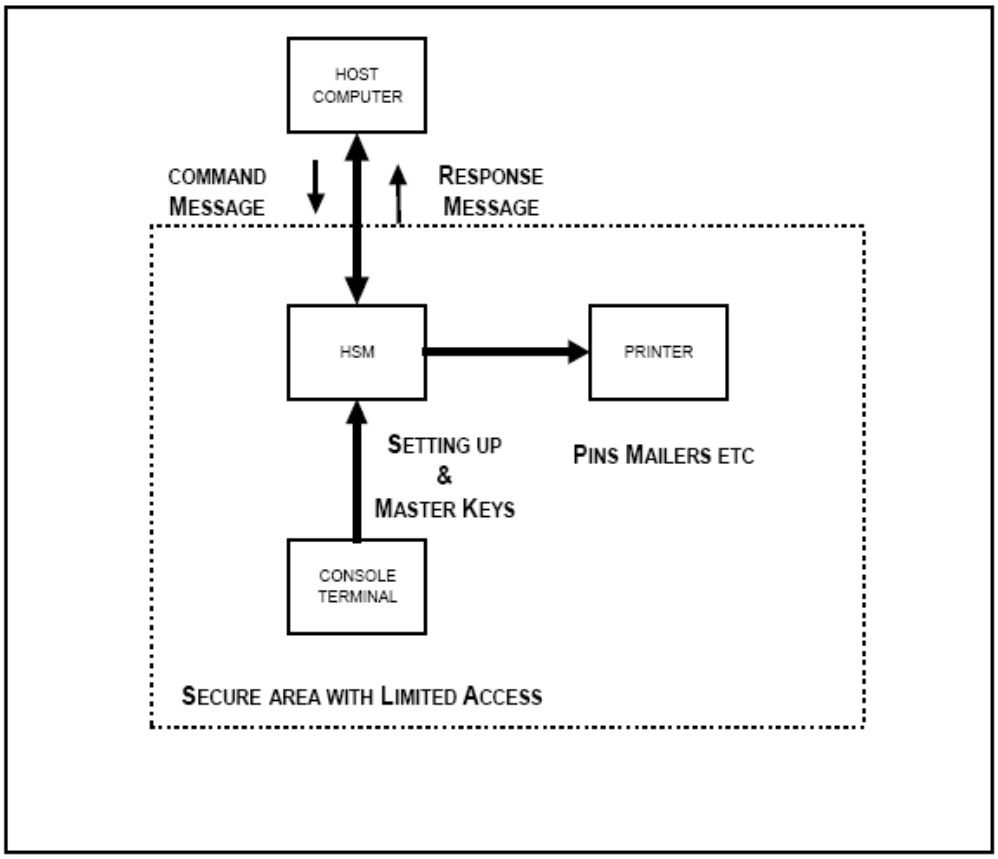


Figure 6: HSM in a Typical System

2.1.1. Preparing the HSM for use

2.1.1.1. Connecting power

A power cable suitable for the local mains connectors is supplied with each HSM. Plug the cable into the IEC style connector on the power module at the rear of the HSM. There is no on/off switch on the HSM, so it is ready to operate when it is connected and power is turned-on.

2.1.1.2. Booting up

If a Console terminal is connected to the HSM when power is applied or when the HSM is reset using the RESET button on the front panel, the following booting up message is displayed:

```
The HSM is booting up, please wait...
Booting complete.
Secure>
```

If the booting up message includes error messages (an example is shown below), these can be ignored.

```
The HSM is booting up, please wait...
LMK File read error
Old LMK File read error
Booting complete.
Secure>
```

In addition to such messages, the booting up message may be preceded by a few spurious characters. These can also be ignored.

2.1.2. Configuring security

The security configuration of the HSM and some processing parameters are set by the `CS` (Configure Security) console command; the settings can be examined by the `QS` (Query Security) console command. See the *HSM8000 Console Reference Manual* for full details of these commands.

To configure the HSM security settings

1. Ensure the HSM is offline. Then enter `CS` at the HSM console.
2. When prompted, enter the following:

Field	Description
PIN length	Length of the PIN. Enter `04`.
Encrypted PIN length	Length of the PIN to be stored as an encrypted PIN under the LMK. Enter `05`, as the encrypted PIN is one character greater than the length of the PIN.
Echo	Whether or not to display passwords and other secret values on the console as they are entered. Enter `F` as displaying passwords in this way is a security risk.
Atalla ZMK variant support	Whether or not to enable optional Atalla variants within commands. Enter `F`, as NOMAD CORTEX doesn't support Atalla ATMs.
Racal or Australian transaction key	Which variants of transaction key to support. Select `R`, as NOMAD CORTEX uses the RACAL variants.

Field	Description
User storage key length	Length of the keys stored in User storage. If the system has not yet migrated to 3DES, enter `S`. If the system has migrated to 3DES, enter `D`.
Select clear PINs	Whether or not to enable the clear PIN support command NG. Enter `N` as this is a security risk.
Enable ZMK translate command	Whether or not to enable translation of ZMKs from under another ZMK. Enter `N` as this is a security risk.
Enable X9.17 for import -	Whether or not to enable support for the ANSI X9.17 mechanism for key import. Enter `Y` if the system has migrated to 3DES.
Enable X9.17 for export	Whether or not to enable support for the ANSI X9.17 mechanism for key export. Enter `Y` if the system has migrated to 3DES.
Solicitation batch size	Minimum size for batches of returned PIN solicitation mailers. NOMAD CORTEX doesn't support such mailers. Enter `1023`.
Disable single-DES	For the HSM8xxx series only. When using 3DES, whether or not to prevent the use of single DES. Enter `N` as NOMAD CORTEX needs to maintain backwards compatibility with single-DES keys.
ZMK length	The length of the ZMK, single or double. If the system has not yet migrated to 3DES, enter `S`. If the system has migrated to 3DES, enter `D`.
Enable decimalisation table checks	For the HSM8xxx series only. Whether or not to disable restriction of values in the decimalisation tables, used for deriving and verifying PIN offset values Enter `N` as NOMAD CORTEX doesn't make use of decimalisation tables.
PIN encryption algorithm	PIN encryption algorithm to be used when encrypted PINs are stored by the card issuer. Enter `A` as NOMAD CORTEX uses the Visa method.

Field	Description
Card/Password Authorisation	Method of authenticating security officers requesting a security state change. The Authorised State is a condition the HSM can be placed in for sensitive data processing. This authorised mode is required when input functions at the console or host use clear text data such as key components or unencrypted PINs. Authorised mode can be used in both Online and Offline host states and requires the authorising officers to invoke the higher security level. Before the authorised state can be set the authorising officers need to be verified by the HSM. Officer verification is done by checking either a Smartcard and PIN or a password (16 alphanumeric characters.) If the Password option is not set when the LMK is created, the password option will not be available as no password is created and stored with the LMK components. The verification method is to be decided by the bank, although use of smart cards is more usual. Enter `C` if you want to use smart cards Enter `P` if you want to use passwords
Card issuer password	This option enables Thales to change the card issuer password for LMK and authorising officer Smartcards. If the password has been changed you will be advised when new cards are delivered. **Do not enter anything here**. If this setting is changed, it will not be possible to format the smart cards using the `FC` command.

2.1.3. Connecting to the Console terminal

The Console interface is a configurable serial port provided on both the front and rear panels of the HSM. It is the right-hand connector on the rear panel. The connector is an industry standard D-type 9-way female connector (socket) with screw fittings. Note that the two Console port connections are electrically the same and only allow control by connection to one at a time.

The Console is required during installation, and for operations in which secret data is entered into the HSM. Console operations include generating and loading the Local Master Keys (LMKs) and passwords, putting the HSM into the Authorised state, generating manually-distributed master keys and performing diagnostic functions. The terminal must therefore be located in a secure access area.

The Console terminal is not supplied with the HSM and must be provided by the User. It can be any type of standard terminal, e.g. a VT100. It is connected to the HSM by a cable, also User supplied, which must not be more than 50ft (1524cm) in length. The Console terminal is not required for normal day-to-day HSM operation, so a single terminal can be shared across a set of HSMs.

2.1.3.1. Console specification

The Console settings for NOMAD CORTEX are:

- ASCII character set
- RS-232-C interface
- 9600 baud rate
- 8 data bits, 1 stop bit, no parity
- XON/XOFF flow control

The Console must not be able to store information and display it at a later time (because some data may be of a sensitive nature).

Character transmission rates and formats are specified by the User and can be configured at the time of HSM installation. The Console must be capable of operating at the HSM factory default settings. See the *HSM8000 Security Operations Manual*. RTS must be asserted to allow output from the HSM.

2.1.3.2. Confirming correct Console configuration

Assuming the HSM default settings (as shipped from the factory, or after a cold start) apply, configure the Console as instructed in the *HSM8000 Security Operations Manual*, and for full duplex with no local echo.

Press the <Return> key. The HSM should respond with:

```
Online >
```

which indicates that correct communications have been achieved but a valid command has not been entered.

2.1.3.3. Console and Auxiliary port interface signals

Pin	Signal	Details
1	Protect Ground	Connected to the HSM chassis
2	TX Data	To HSM
3	RX Data	From HSM
4	RTS (Request To Send)	To HSM (must be asserted)
5	CTS (Clear To Send)	From HSM (always asserted)
6	DSR (Data Set Ready)	From HSM (always asserted)
7	Signal Ground	—
8	DCD (Data Carrier Detect)	From HSM (always asserted)
15	TX Clock (DCE Source)	16 x Baud from HSM
17	RX Clock (DCE Source)	16 x Baud from HSM
20	DTR (Data Terminal Ready)	To HSM (ignored)

2.1.4. Connecting to the printer

A printer can be connected to allow the HSM to print PIN mailers or generate and print components of manually-distributed keys. A parallel or serial printer can be used.

The printer port is used to drive a standard parallel printer (not a “Win-printer” type) implementing compatibility, nibble and byte mode data transfer. The interface is a 25-way ‘D’ type female connector (printer port) on the rear panel of the HSM. There are no configuration parameters, but the HSM is set for parallel printing using the Configure Printer (CP) console command.

The auxiliary interface is used to drive a serial printer. The interface is a 25 way ‘D’ type plug (serial port) on the rear panel of the HSM. The auxiliary port has the same settable configuration parameters as the Console port and is set using the console commands Configure Auxiliary (CA) or Configure Printer (CP).

2.1.4.1. Serial printer specification

Character set	ASCII
Interface	RS-232-C (DCE) serial data (not parallel)
Baud	300 to 38400 bps (default 300 bps)
Stop bits	1
Data bits	7 or 8 (default 7)
Parity	Odd, even or none (default odd)
Flow control	XON, XOFF

It is recommended that printers are located in a secure access-controlled area.

2.1.5. Connecting to the Host

Normally, HSMs interface to NOMAD CORTEX via TCP/IP. If the bank has purchased a low-speed HSM for PIN printing, it connects to NOMAD CORTEX via an asynchronous connection. The following figure shows the pinout for an asynchronous connection.

2.1.5.1. Host port connections for asynchronous connection

Pin	Signal	ASCII asynchronous, DCE
1	Protect Ground	Connected to the HSM chassis
2	TX Data	To HSM
3	RX Data	From HSM
4	RTS (Request To Send)	To HSM (ignored)
5	CTS (Clear To Send)	From HSM (always asserted)
6	DSR (Data Set Ready)	From HSM (always asserted)
7	Signal Ground	—
8	DCD (Data Carrier Detect)	From HSM (always asserted)
15	TX Clock (DCE Source)	16 x Baud from HSM
17	RX Clock (DCE Source)	16 x Baud from HSM
20	DTR (Data Terminal Ready)	To HSM (ignored)
23	Data Rate Select	Not connected
24	TX Clock	To HSM (Ignored)

2.2. Configuring the HSM

2.2.1. General

This section describes how to physically configure the HSM to work with the Host system.

Configuration of the alarms and the security settings can be found in the *HSM8000 Security Operations Manual*.

Entry of commands and data at the Console is not case sensitive (i.e., `A` has the same effect as `a`).

Spaces can be inserted between characters to ease legibility during entry; they are ignored by the HSM. However they cannot be used between command characters (e.g. the `LK` command cannot be successfully entered as `L K`).

When entering sensitive (clear text) data, use the Inhibit Echo Back facility to ensure that the HSM does not echo the data to the Console screen. This is set at configuration using the `CS` (Configure Security) command. Instead of displaying the data, the HSM displays a star for each character entered. Thus:

```
0123456789ABCDEF
```

is shown on the screen as:

```
****************
```

To exit from a command during data entry, press `<Control>` and `C` simultaneously. The HSM responds with:

```
TERMINATED
```

2.2.2. Configuring the Printer port

A printer is required to print PIN mailers or generate and print components of manually distributed keys. The HSM is configured from the Console to use either a parallel printer connected to the printer port or a serial printer connected to the auxiliary port.

The Printer port is a standard ECP/EPP parallel port with a standard 25-pin D-type connector.

The Printer port implements compatibility, nibble and byte mode data transfer; therefore a “Win-printer” that requires the installation of specific software drivers cannot be used. In normal operation, the HSM is set into the Authorised state by the use of the Console. Then printing can start. The printer must be located in a secure access-controlled area.

The Printer port can be configured while the HSM is in online, offline or secure state. The User is invited to select either serial printer (connected to the auxiliary port) or parallel printer (connected to the printer port). If Parallel printer is selected then there are no other parameters. If Serial printer is selected then the settings for the auxiliary port must be selected.

Note that the Auxiliary port can also be configured using the `CA` (Configure Auxiliary) command; any previous settings are overwritten.

Example 1: (serial printer)

```
CP <Return>
Series/Parallel? [S/P]: S <Return>
BAUD RATES WORD FORMATS
1. 300 1. 7 bits, no parity, 1 stop
2. 600 2. 7 bits, odd parity, 1 stop
3. 1200 3. 7 bits, even parity, 1 stop
4. 2400 4. 8 bits, no parity, 1 stop
5. 4800 5. 8 bits, odd parity, 1 stop
6. 9600 6. 8 bits, even parity, 1 stop
7. 19200
8. 38400

In response to the prompts, enter the number of the
desired option or <RETURN> for no change.

Printer port baud rate: (current value = 6) <Return>
Printer port word format: (current value = 4) 3 <Return>
Print test page? [Y/N]:
```

The values entered take effect immediately after the command has been completed.

Example 2: (parallel printer)

```
CP <Return>
Series/Parallel? [S/P]: P <Return>
Print test page? [Y/N]:
```

2.2.2.1. Default printer settings

The default settings for a serial printer are:

Character set	ASCII
Interface	RS-232-C (DCE) serial data (not parallel)
Baud	9600
Stop bits	1
Data bits	8
Parity	None
Flow control	XON, XOFF

Note that the parallel Printer port has no configurable parameters.

2.2.3. Configuring the Auxiliary port

The Auxiliary port is configured as a piece of Data Communications Equipment (DCE). The Auxiliary port can be used to connect a serial printer or other serial device. It is configured from the Console, either as part of the printer configuration or individually. The default settings are the same as the Console port. The baud rate and word format can be changed to any convenient value.

As for parallel printing, the HSM is set into the Authorised state for printing by the use of the Console. The printer must be located in a secure access-controlled area.

The Auxiliary port can be configured while the HSM is in online, offline or secure state.

The variables to be configured for NOMAD CORTEX are as follows:

Baud	9600
Stop bits	1
Data bits	8
Parity	None
Flow control	XON, XOFF

The procedure is similar to configuring the Console port. The following is an example of configuring the Auxiliary port for 9600 baud, 7 data bits, even parity and 1 stop bit.

Example:

```
CA <Return>
BAUD RATES WORD FORMATS
1. 300 1. 7 bits, no parity, 1 stop
2. 600 2. 7 bits, odd parity, 1 stop
3. 1200 3. 7 bits, even parity, 1 stop
4. 2400 4. 8 bits, no parity, 1 stop
5. 4800 5. 8 bits, odd parity, 1 stop
6. 9600 6. 8 bits, even parity, 1 stop
7. 19200
8. 38400

In response to the prompts, enter the number of the
desired option or < RETURN > for no change.

Auxiliary port baud rate: (current value = 6) <Return>
Auxiliary port word format: (current value = 4) 3
<Return>
```

The values entered take effect immediately after the command has been completed.

2.2.3.1. Default settings (Auxiliary port)

The default settings for the Auxiliary port are:

Baud	9600
Stop bits	1
Data bits	8
Parity	None

2.2.4. Configuring the Host port

The HSM Host interface can be configured using the Console to emulate a number of types of data communications equipment and control equipment. At the end of the configuration, the User is given the option to save the host interface settings to a Smartcard.

To configure the Host port, set the HSM into the offline state (insert the key in one of the KEY switches on the front panel and rotate it a half turn), with power applied and the Console terminal connected.

The Console displays:

```
HSM going OFFLINE, press Reset to go Online.
Offline>
```

Enter `CH <Return>` (Configure Host) to initiate the set-up dialogue. Examples of the various configurations are given below. User inputs are shown underlined. The `CH` command can also be used while the HSM is in the secure state.

- For the HSM7400, you need to set up asynchronous emulation (see *2.2.4.1 Setting up Asynchronous Emulation*)
- For HSM7100, HSM7110 and the HSM8xxx series, you need to set up TCP/IP (see *2.2.4.2 Setting up Ethernet communications*)

2.2.4.1. Setting up Asynchronous Emulation

In Asynchronous Emulation the HSM is viewed by the Host as a DCE (Data Communications Equipment) device, and does not require a modem. The electrical interface between the Host and the HSM can be configured to conform to the RS-232, V.35 or X.21 standard.

To configure the HSM for asynchronous communications:

- The port must be defined as a DCE
- Details held in software must be configured

The following variables can be configured:

- The required length of the message header. This is normally set to 4, but can be set between 1 and 255. This depends on the value used by the Host computer application.
- The baud rate of the Host computer port
- The word format of the Host computer port
- The required communications protocol, either standard or transparent asynchronous ASCII
- The asynchronous terminating characters. The terminating sequence can be either one or two characters. To select the terminating characters four hexadecimal values must be entered. If only one terminating character is required, enter the first two hexadecimal values followed by 00.

Message Header length

Each transaction to the HSM begins with a string of characters (header) which the Host can use to identify the transaction (or for any other purpose). The HSM returns the string unchanged to the Host in the response message. The length of the header can be set to any value between 1 and 255; the default value is 4, and NOMAD CORTEX uses 1.

Transparent asynchronous communications

In the standard asynchronous mode of communication, codes like STX (X'02) and ETX (X'03) have a special meaning, but they can sometimes occur in a stream of binary data, where that special meaning does not apply.

To avoid ambiguity, Transparent Asynchronous Communications mode is used. This has a simplified message format (for details see the HSM Programmers Manual).

The Host port of the HSM must be configured for Transparent Asynchronous Communications and 8-bit data transfers.

NOMAD CORTEX requires transparent asynchronous communications as EMV and MAC messages are used. The variables to be configured for NOMAD CORTEX are as follows:

Character set	ASCII
Baud	9600
Stop bits	1
Data bits	8
Parity	None
Message header length	1
Transparent	Y
Terminating characters	03 00

In the example, transparent asynchronous communications is enabled and the message header length is set to 1 character. The Host baud is changed to 9600 bps and the word format is set to 8 data bits, no parity and 1 stop bit. No option of message terminating characters is offered because this is fixed (value X'03, "ETX" character).

Example:

```
Offline> CH <Return>
Message header length (1-255): 1
Asynch, Ethernet, or SNA-SDLC 3274 [A/E/S]: A
Transparent mode (Y/N): Y
Interface: DTE or DCE [T/C]: C
BAUD RATES WORD FORMATS
1. 300   1. 7 bits, no parity, 1 stop
2. 600   2. 7 bits, odd parity, 1 stop
3. 1200  3. 7 bits, even parity, 1 stop
4. 2400  4. 8 bits, no parity, 1 stop
5. 4800  5. 8 bits, odd parity, 1 stop
6. 9600  6. 8 bits, even parity, 1 stop
7. 19200
8. 38400
In response to the prompts, enter the number of the
desired option or <RETURN> for no change.

Host baud rate: (current value = 1) 6 <Return>
Host word format: (current value = 2) 4 <Return>
Save HOST settings to smart card? [Y/N]: N <Return>
```

After completing the procedure, reset the HSM by pressing the RESET button on the front panel. The new values do not take effect until this has been done.

To specify use of Transparent Asynchronous mode, you will also need to modify the optional flag TASYNC in the ENCRYPT subsection of the NOMAD CORTEX-specific configuration file ctxcfg to read `TRUE`.

2.2.4.2. Setting up Ethernet communications

The HSM can be set to use either Ethernet interface 10 base-T or 100 base-TX.

Software parameters

There are a number of prompts for configuring the software:

- The message header length
- The character set (either ASCII or EBCDIC)
- The availability of a UDP port
- The availability and number of TCP ports. The number of TCP/IP sockets available has a maximum of 8 but is reduced to 7 if UDP protocol is enabled.
- The Keep_Alive timer, which enables TCP to periodically check whether the other end of a connection is still open. This enables the HSM to free resources by closing any unused connections.
- The IP address, which is the Internet Protocol address of the unit in the system. It is four decimal numbers, each not exceeding 255_{10}.
- The Well-Known-Port address, which is the published TCP Port address of the HSM, in the range 00000_{10} to 65535_{10} representing an address in the range 0000_{16} to $FFFF_{16}$.
- The default gateway address, which is the Internet Protocol address of the default gateway in the system. It is four decimal numbers, each not exceeding 255_{10}.
- The subnet mask, which is used to define the network class.

The addresses have no cold start defaults; if `< Return >` is entered after a prompt, the previously-configured address is used.

TCP configuration is given in the following example.

Example: (Ethernet communication with TCP/IP)

```
Offline> CH <Return>
Message header length (1-255): 4
Asynch, Ethernet, or SNA-SDLC 3274 [A/E/S]: E
EBCDIC/ASCII (E/A): A
Enter IP address: 128.100.3.1
Enter Well-Known-Port address: 01500
Enable UDP? [Y/N]: N
Enable TCP? [Y/N]: Y
Number of connections? [1 - 64]: 6
Enter TCP Keep_Alive timeout (1-120): 120
Enter default gateway address: 128.100.3.5
Enter subnet mask: 255.255.255.000
Save HOST settings to smart card? [Y/N]: N <Return>
```

After completing the procedure, reset the HSM by pressing the RESET button on the front panel. The new values do not take effect until this has been done. To read the current configuration, use command `QH`:

```
Online> QH <Return>
Message header length: 04
Protocol: Ethernet
Character format: ASCII
IP address: 128.100.003.001
Well-Known-Port address: 01500
Transport: TCP, 6 connections
TCP keep_alive value: 120 minutes
Default gateway address: 128.100.003.005
Subnet mask: 255.255.255.000
```

2.3. Initialising the HSM

The LMKs must be loaded into the HSM before it can be used. Other keys can optionally be input into the HSM directly, or via a NOMAD CORTEX interface, once the LMK is stored.

The components of the key are combined by means of the exclusive-OR operation within the HSM. A key check value is returned for each component entered, as well as for the full combined LMK.

Once the key components have been saved to smart cards, it is necessary to load the LMK components back into the HSM to form the final LMK encryption key. You will also need to load LMKs from smart cards if:

- the HSM has been opened (for security purposes, this erases the LMKs)
- there has been a power failure, and the back up battery has expired
- the HSM has been disturbed, causing the motion sensor to activate

The stored LMK value can be checked and verified periodically as a security measure. Similarly, the original smart cards and any duplicates should also be checked over time.

The figure below provides an overview of the LMK generation process:

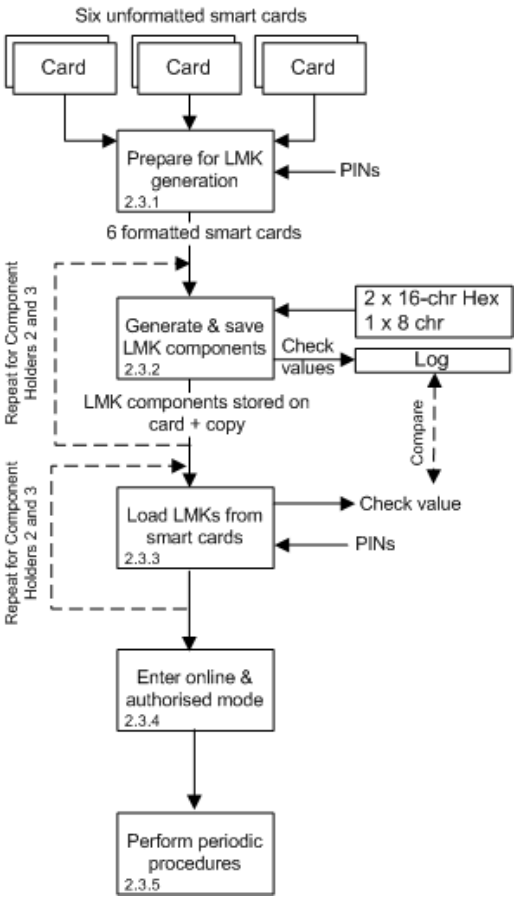


Figure 7: Generating LMKs

2.3.1. Preparing for LMK generation

For these procedures you will need:

- access to the HSM, both physically (i.e., by obtaining keys for the front and rear of the device) and via a Console
- six smart cards, with blank identification labels affixed
- three *Component Holders*—trusted officers of the bank who will be responsible for looking after the six smart cards

To prepare the HSM for LMK generation

1. Gather the Component Holders together at the HSM (or nearby)
2. Read the safety warnings relating to access inside the machine. These are at the front of the HSM operation manual.
3. Start the HSM (boot mode is *Online*, *Non-Authorised* mode)
4. Connect the HSM to a suitable monitor and provide unimpeded access to the HSM unit
5. Switch the HSM to offline mode. To do this, turn the key on the rear of the unit one quarter turn clockwise, then release it and allow it to return to the upright position. This mode retains the current authorisation state; if this is already *authorised* then return to the non-authorised state by entering `C` at the prompt.
6. Ensure that the HSM is in echo on mode.

To format the Smart cards

1. Gather the Component Holders together at the HSM (or nearby)
2. Give each Component Holder two smart cards with blank labels
3. Ensure that the HSM is in the *Online*, *Offline* or *Secure* state. Then, at the HSM console, enter `FC`.
4. Component Holder 1 inserts a smart card
5. Component Holder 1 selects a secret PIN for the card, enters this once then re-enters to confirm.
6. Component Holder 1 writes down the PIN twice. One copy is kept in a sealed envelope in a safe, and another copy in another, similarly secure location.
7. Enter the time in hhmmss format
8. Enter the date in yymmdd format
9. Enter the Issuer ID: Name of issuing organisation
10. Enter the User ID: Name of User (owner) of the card
11. Component Holder 1 repeats steps 4-10 for the second card
12. Component Holders 2 and 3 then carry out steps 4-10 for their two cards, resulting in a total of six formatted cards.

2.3.2. Generating and saving LMK components

For this procedure, each of the three Component Holders will need to invent two 16-character hexadecimal numbers (*components*) to form the LMK. Instruct the Component Holders to keep these numbers secret, thus ensuring that at no time does any one Component Holder have access to all the LMK components.

Also required are:

- a log for recording LMK check values. The bank typically creates this document according to its needs. Consult the relevant authority for the precise procedure at your bank.
- The six formatted smart cards created in *To format the Smart cards*

To generate LMK components and save them to smart cards

Note: To ensure maximum security, it is essential that the Component Holders do not disclose the hexadecimal numbers that they have created, or their smart card's PIN to anyone else

1. Gather the Component Holders together at the HSM (or nearby)
2. Ensure the HSM is in the *Secure* state. Then, at the HSM console enter `GK`.
3. Enter `y` to proceed using physical keys
4. Pull the arming ring on the rear of the unit
5. Open the front panel and set the (spring-like) switch to the left position
6. Shut the front of the unit
7. Select the LMK component to enter. Enter `1` to enter component 1.
8. Component Holder 1 enters the first 16-character hexadecimal number, followed by the second.
9. Component Holder 1 enters a third character set for added security. Typically, the date in ddmmyyyy format is used. The HSM requests a smart card onto which to copy the LMK component set.
10. Component Holder 1 inserts the first smart card and enters the PIN for the card. The Holder then presses `<Enter>`. The HSM displays a check value for that component.
11. Component Holder 1 records the check value in the LMK log, and then labels the smart card 'LMK component 1'.
12. Component Holder 1 enters `<y>` to make an additional copy of the component on another smart card.

Note: It is advisable to create duplicate copies of the components. However, bear in mind the security implications of having many copies.

13. Component Holder 2 performs steps 7-12. In step 7, when prompted to select the LMK component to enter, the Component Holder enters `2`.
14. Component Holder 3 performs steps 7-12. In step 7, when prompted to select the LMK component to enter, the Component Holder enters `3`.

2.3.3. Loading LMKs from the smart cards

For this procedure, you will need:

- the three Component Holders, each with their two formatted smart cards
- the LMK log

To load LMKs from the smart cards

Note: To ensure maximum security, it is essential that the Component Holders do not disclose their smart card's PIN to anyone else.

1. Gather the Component Holders together at the HSM (or nearby)
2. Ensure that the HSM is in the *Secure* state. Then, at the HSM console enter `LK`.
3. Accept the warning prompt
4. Component Holder 1 inserts the first component card and presses `<Enter>`. The Holder then enters the PIN in response to the prompt and presses `<Enter>` again.
5. The HSM reads the card and returns the check value, before prompting for further component cards. The check value should correspond to that returned in step 10 of *To generate LMK components and save them to smart cards*.
6. Component Holder 2 performs steps 4-5, followed by Component Holder 3.
7. Make a note of the final check value. This is most important. At each stage, the check value is the resultant value from all the cards entered so far, thus the resultant of all components is the check value that will be used for future verification of the LMK.
8. Enter `n` to finish the reading process when all components have been entered.
9. Arm the HSM unit by pulling the tamper-protection ring on the rear panel. Confirm the armed light on the front panel is illuminated.

   The smart cards will now need to be stored in separate secure locations. If LMK components have to be re-entered at a later date, it is often found that one or more of the Component Holders cannot be found. For this reason banks typically try to arrange matters so that the smart cards are distributed among the Component Holders in such a way that not all of them need to be present.

   For example, if there are three smart cards (C1, C2, C3), and three Component Holders (H1, H2, H3), then H1 could have (C2, C3), H2 could have (C1, C3), and H3 could have (C1, C2). In this arrangement any two out of the three Component Holders would be sufficient.

2.3.4. Entering online and authorised mode

After generating the LMKs and storing them from smart cards, you need to return the unit to *online* mode and check it is set to *authorised*. This mode is necessary for entering other encryption keys and generating PINs.

To enter online and authorised mode

1. Press the Reset button in the centre of the lower rear panel on the HSM. The screen displays the prompt `Online >`.
2. At the HSM console, enter `A` (for authorised mode).
3. Insert the first LMK component smart card and enter its PIN. Insert the second LMK component smart card and enter its PIN.
4. If authorisation was successful, the prompt becomes:

   `Online - AUTH >`.

   If authorisation was unsuccessful, this is notified and the prompt remains `Online >`. In this case, enter `A` and retry.
5. If authorised mode is not required, enter `C` to cancel and return to not authorised mode, ready for normal operation.

2.3.5. Performing periodic procedures

The following procedures should be carried out periodically.

To check the LMK value stored on the HSM

1. Ensure that the HSM is in the *Online*, *Offline* or *Secure* state. Then at the HSM console, enter `V`.
2. The HSM carries out an internal check and displays a Master Key check value
3. Verify that this agrees with that noted after step 7 of *To load LMKs from the smart cards*

To check smart card content agrees with the stored LMK

1. Insert the card that you wish to check into the HSM card reader
2. Enter `VC` at the HSM console
3. The HSM carries out a check of the smart card and displays a check value, indicating whether the card passed or failed the check

To change LMKs

For this procedure, you will need:

- the three Component Holders, each with their two formatted smart cards
- the LMK log

You should change the LMKs at least every two years. To do this, you first move the old LMKs into secure key storage within the HSM, and then generate new LMKs.

1. Ensure that the HSM is in the *Secure Authorised* state. Then, at the HSM Console, enter `LO`.
2. Component Holder 1 inserts the smart card containing the first LMK component into the HSM card reader, and enters the PIN when prompted.
3. The HSM displays a check value, which should correspond to that noted in the LMK log. The HSM prompts for more components. Enter `Y`.
4. Component Holder 2 performs steps 2-3, followed by Component Holder 3.
5. Once all the old LMKs have been moved into key storage, generate and store new LMKs by following the procedures described in sections 2.3.1 to 2.3.4.

3. Managing Encryption Keys

You use the following procedures to generate and update the encryption keys used to guarantee the security of transactions. You should only carry out changes to the encryption keys if you have received the appropriate training. Keys are normally set up during installation by NOMAD specialists—and once set up, these keys will not often need to be changed.

The figure below illustrates the main stages in setting up the NOMAD CORTEX encryption system (the relevant section is indicated in the bottom left of each box).

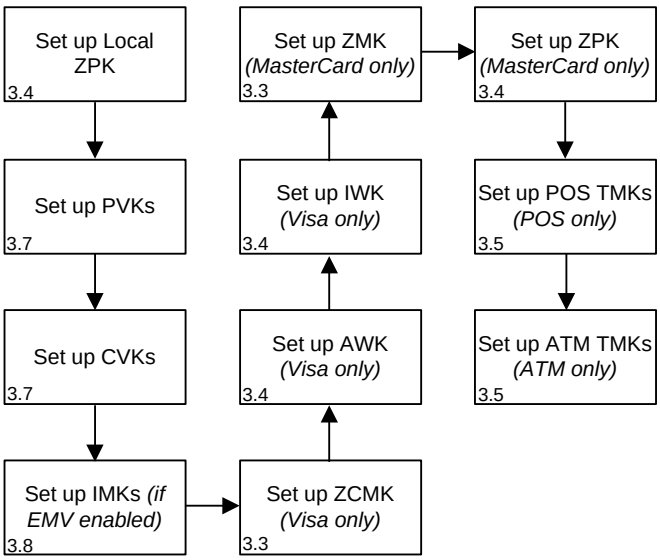


Figure 8: Setting up the encryption system

3.1. Maintaining keys encrypted under an LMK

For a full description of the function of LMKs, see page 19. NOMAD CORTEX enables you to view, and in some cases, modify the expiry date for keys stored under the LMK, such as those used for MAC'ing messages.

To generate a key encrypted under the LMK

1. At the HSM console enter `KG`.
2. When prompted, enter the following:

Field	Description
Key length	Length of the key to generate as follows: 1= Single length 2= Double length 3= Triple length (not currently used by CORTEX)
Key type	This is three digits. The first digit is the key variant as follows: ZPK, PVK, TPK, TMK, TAK = 0 ZMK, ACMK, SMIMK = 1 SMCMK = 2 CVK = 4 DACMK = 6 DNMK = 7 The second/third digits are the LMK key pair to use: ZMK (04-05) = 00 ZPK (06-07) = 01 TMK, TPK, PVK, CVK (all 14-15) = 02 MAC (16-17) = 03 IWK/AWK (24-25) = 07 IMKs (28-29) = 09
Key scheme (LMK)	The LMK scheme. This is the type of encryption that the HSM uses for the LMK: Z= Single length DES key encrypted using ANSI X9.17 methods, U= Encryption of a double-length key using variant method. Used for encryption of keys under LMK and can be used for import and export of keys. T= Encryption of a triple length key using variant

Field	Description
	method. Used for encryption of keys under LMK and can be used for import and export of keys. X= Encryption of a double-length key using ANSI X9.17 methods only available for import and export of keys. Y= Encryption of a triple length key using ANSI X9.17 methods only available for import and export of keys.
Key scheme (ZMK)	The ZMK scheme. This is the type of encryption that the HSM uses for the ZMK. See Key Scheme (LMK) above for values.
ZMK *[optional]*	ZMK encrypted under LMK pair 04-05 (as generated using the procedure described in *To add a ZMK* to CORTEX on page 62). Enter the value you specified for Key scheme (LMK) followed by the encrypted ZMK, for example: `U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX` Or, press `<Return>` at this prompt to return *only* the key encrypted under the LMK.
ZMK key check value *[optional]*	ZMK key check value (as generated using the procedure described in *To add a ZMK* to CORTEX on page 62) Press `<Return>` at this prompt and the test will not be carried out.
ZMK variant *[optional]*	Used only when interworking with Atalla systems. Press `<Return>` at this prompt.

3. The HSM returns three sequences of characters:

Field	Description
Key under LMK	Key encrypted under the LMK for local storage
Key under ZMK *[optional]*	Key encrypted under the ZMK for transmission
Key check value	Check value for the key

Make a confidential note of the sequences. Now you need to enter the key into NOMAD CORTEX (see *To add a previously generated key encrypted under the LMK* on page 54.)

To add a previously generated key encrypted under the LMK

1. From the menu bar select **Core>Key Maintenance>LMK Maintenance>Add**. The **Encrypt keys under LMK** screen appears displaying default details.
2. Complete the fields as follows:

Field	Description	M	O
Key type	Type of NOMAD CORTEX encryption key	✓	×
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✓	×
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	✓	×
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	×
Status	Status of the key	✓	×
Expiry date	Default expiry date of a NOMAD CORTEX encryption key	✓	×
Key length	Whether this is a single or double-length encryption key	✓	×
Key value	Value of the encryption key, encrypted under the LMK	✓	×
Check value	Check value of the encryption key	✓	×
Activation date	Date the encryption key was activated	×	×
Activation time	Time the encryption key was activated	×	×

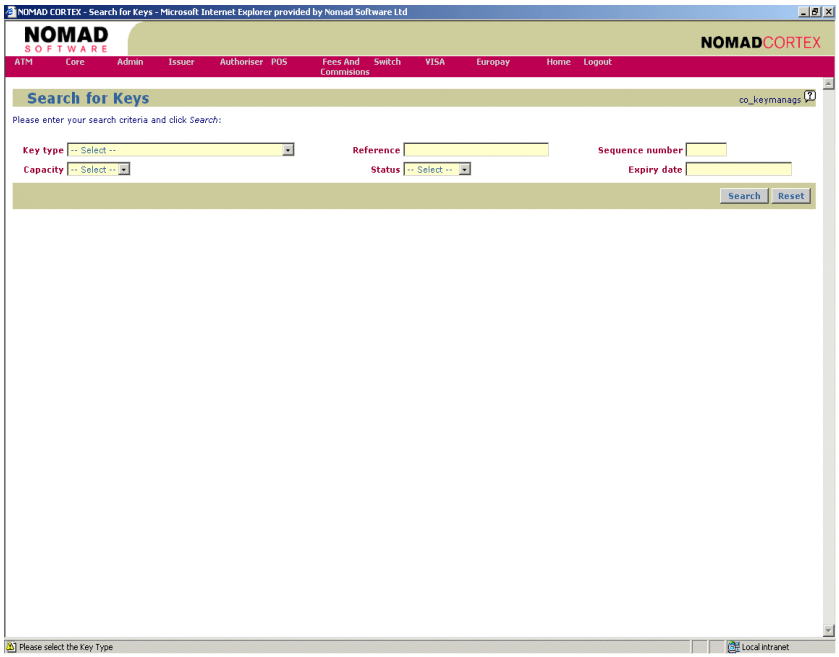
3. When you have finished, do one of the following:
   - ◊ Click **Add** to add the encryption key to the database. If the addition is successful, a confirmation message appears, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values

3.2. Viewing the status of encryption keys

NOMAD CORTEX enables you to review the statuses of all encryption keys currently stored in the database.

To view encryption key statuses

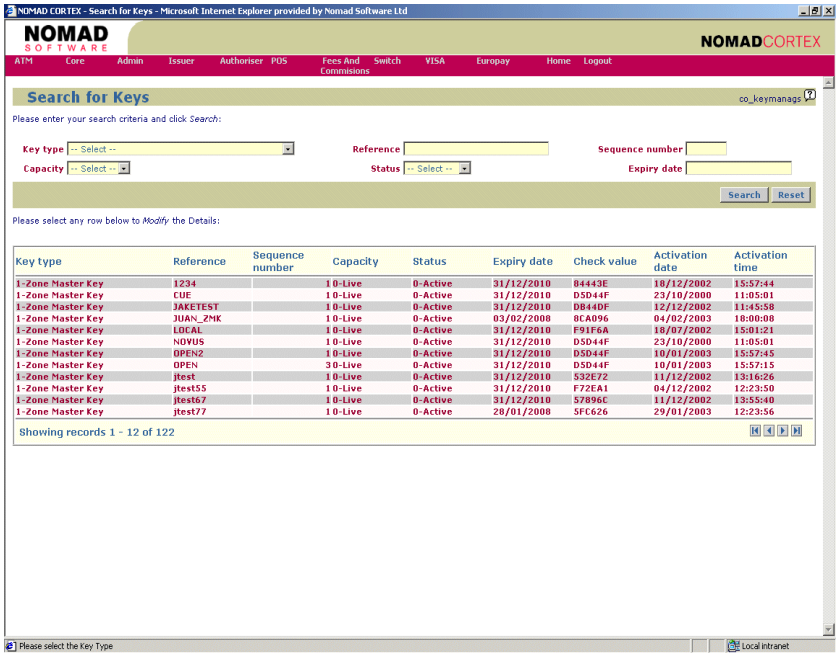
1. From the menu bar, select **Core>Key Maintenance>Key Management>Search**. The **Search for Keys** screen appears:



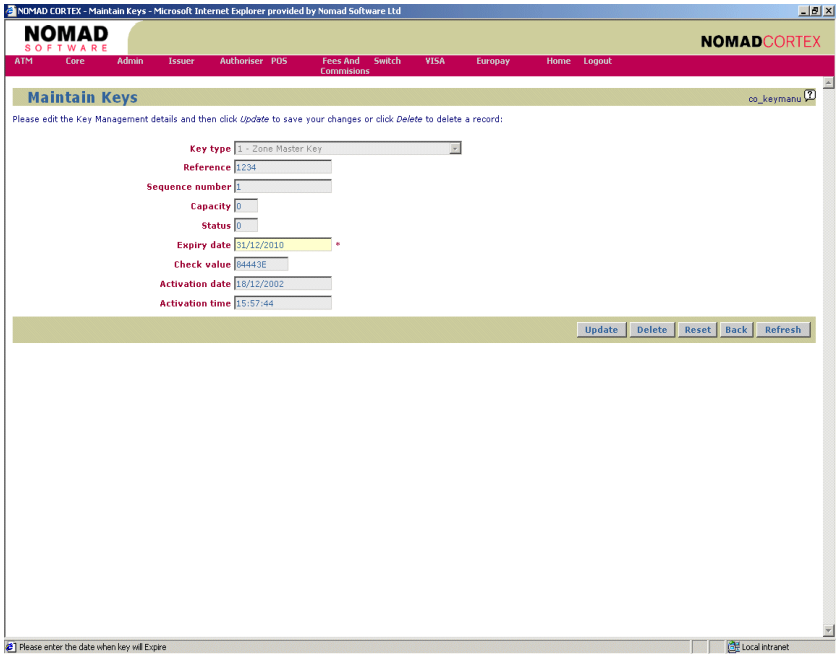
2. In this screen, you can search for a specific key or group of keys using the following fields:

Field	Description	M	O
Key type	Type of encryption key	✓	✓
Reference	Encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✓	✓
Sequence number	Sequence number used to distinguish between CORTEX encryption keys that have the same Key type and Reference	✓	✓
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✓
Status	Status of the key	✓	✓
Expiry date	Default expiry date of a NOMAD CORTEX encryption key	✓	✓

Enter search criteria and click **Search** to view a list of keys matching the criteria, as in the example below:



3. Click on the key that you want to view or for which you want modify the expiry date to display the **Maintain keys** screen for that key:



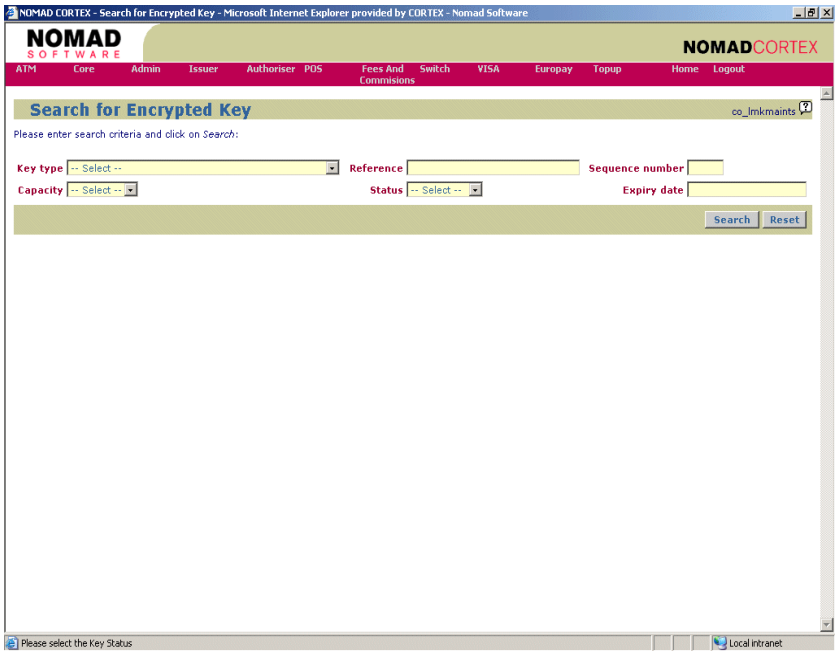
This screen contains the following fields:

Field	Description	M	O
Key type	Type of encryption key	×	×
Reference	Encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	×	×
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	×	×
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	×	×
Status	Status of the key	×	×
Expiry date	Default expiry date of the encryption key	✓	×
Check value	Check value of the encryption key	×	×
Activation date	Date the encryption key was activated	×	×
Activation time	Time the encryption key was activated	×	×

4. When you have finished viewing or modifying the key, do one of the following:
   - ◊ Click **Update** to apply any changes. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Delete** to delete the key and return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes, *OR*,
   - ◊ Click **Refresh** to update the fields of the screen with changes made elsewhere in NOMAD CORTEX

To view or modify encryption keys encrypted under the LMK

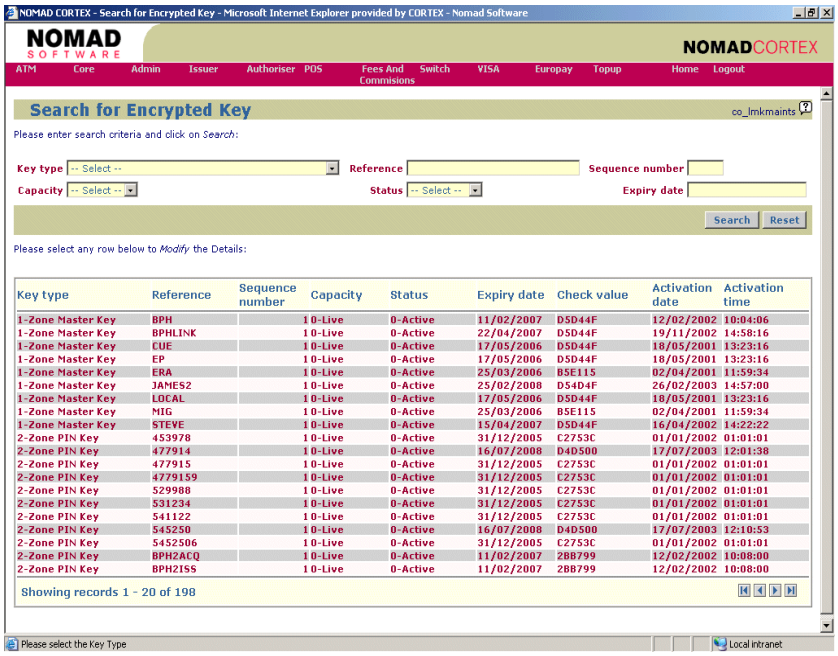
1. From the menu bar, select **Core>Key Maintenance>LMK Maintenance>Search**. The **Search for Encrypted Key** screen appears:



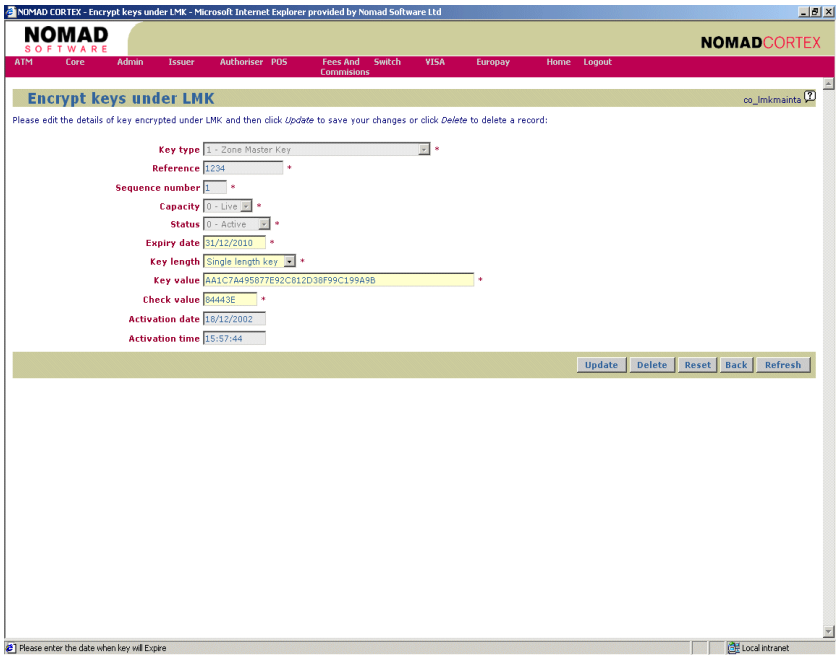
2. In this screen, you can search for a specific key or group of keys using the following fields:

Field	Description	M	O
Key type	Type of NOMAD CORTEX encryption key	✓	✓
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✓	✓
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✓
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	✓	✓
Status	Status of the key	✓	✓
Expiry date	Default expiry date of a NOMAD CORTEX encryption key	✓	✓

Enter search criteria and click **Search** to view a list of keys matching the criteria, as in the example below:



3. Click on the key that you want to view or modify to display the **Encrypt keys under LMK** screen:



This screen contains the following fields:

Field	Description	M	O
Key type	Type of NOMAD CORTEX encryption key	×	×
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	×	×
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	×	×
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	×	×
Status	Status of the key	×	×
Expiry date	Default expiry date of a NOMAD CORTEX encryption key	✓	×
Key length	Whether this is a single or double-length encryption key	✓	×
Key value	Value of the encryption key, encrypted under the LMK	✓	×
Check value	Check value of the encryption key	✓	×
Activation date	Date the encryption key was activated	×	×
Activation time	Time the encryption key was activated	×	×

4. When you have finished viewing or modifying, do one of the following:
   - ◊ Click **Update** to apply any changes. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Delete** to delete the key and return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes, *OR*,
   - ◊ Click **Refresh** to update the fields of the screen with changes made elsewhere in NOMAD CORTEX

3.3. Maintaining ZMKs

For a full description of the function of ZMKs, see page 20.

To encrypt clear ZMK components under the LMK

1. At the HSM console enter `FK`.
2. When prompted, enter the following:

Field	Description
Key length	Length of the key to generate. Enter `2` as the key is double length.
Key type	This is three digits. The first digit is the key variant, and the second/third digits are the LMK key pair to use. Enter `100` to indicate a clear ZMK component to be encrypted under LMK pair (04-05).
Key scheme (LMK)	The LMK scheme. This is the type of encryption that the HSM uses for the LMK. Enter `U` to indicate encryption of a double-length key using variant method.
Component type	Type of component. Enter `X` to indicate XOR.
Number of components	Number of components to be entered this time. Enter `1` as each component is to be entered separately.

3. The HSM returns two sequences of characters:

Field	Description
Encrypted key	Key encrypted under the LMK
Key check value	Check value for the key

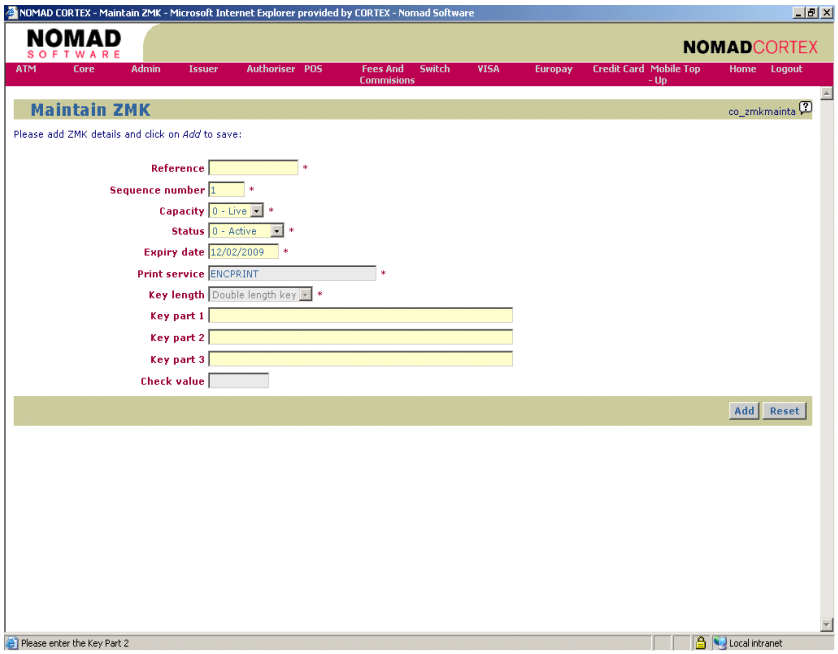
Make a confidential note of the sequences. For Encrypted key, ignore any preceding `U`. Verify that Key check value matches the check value supplied by the third party for the component.

Now repeat steps 1—3 for each of the other two clear ZMK components.

4. You now need to enter the key into NOMAD CORTEX (see *To add a ZMK to CORTEX* on page 62).

To add a ZMK to CORTEX

1. From the menu bar select **Core>Key Maintenance>ZMK Maintenance>Add**. The **Maintain ZMK** screen appears:



2. Complete the fields as follows:

Field	Description	M	O
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type. If you are adding a Visa ZMK (ZCMK) type `VISA`.	✓	✗
Sequence number	Sequence number used to distinguish between ZMKs with the same reference	✓	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✗
Status	Status for the key	✓	✗
Expiry date	Default expiry date of the ZMK	✓	✗
Print service	Print service to which new ZMK will be sent	✗	✗
Key length	Whether this is a single or double-length ZMK	✗	✗

Field	Description	M	O
Key part 1, Key part 2, Key part 3	Enter the three *encrypted* parts of the key. See *To encrypt clear ZMK components under the LMK* on page 61 for details on how to encrypt clear ZMK components.	✓	✓
Check value	Check value of the ZMK	✗	✗

3. When you have finished, do one of the following:

   ◊ Click **Add** to add the ZMK to the database. If the addition is successful, a confirmation message appears.

   If the ZMK is to be exported, send the following to the third party:

   - clear ZMK components
   - check value for each ZMK component
   - check value for the ZMK
   - a note of the order in which the ZMK components where combined on NOMAD CORTEX

   OR,

   ◊ Click **Reset** to return any modified fields to their original values

To generate ZMK components for export

ZMK components need to be generated at the HSM console. However, there is a bug in the HSM firmware that means that the encrypted key output from the HSM terminal `GC` command may not be used with the NOMAD CORTEX screens. So, the following workaround needs to be followed.

- Generate a random clear component using the `GC` command
- Encrypt the clear component using the `FK` command
- Enter the encrypted components into NOMAD CORTEX

1. At the HSM console enter `GC`.
2. When prompted, enter the following:

Field	Description
Key length	Length of the key to generate. Enter `2` as the key is double length.
Key type	This is three digits. The first digit is the key variant, and the second/third digits are the LMK key pair to use. Enter `100` to indicate a clear ZMK component to be encrypted under LMK pair (04-05).
Key scheme	The LMK scheme. This is the type of encryption that the HSM uses for the LMK. Enter `U` to indicate encryption of a double-length key using variant method.
Clear component	Clear component generated by the HSM. Make a careful note of this.
Encrypted component	Encrypted component generated by the HSM.
Key check value	Check value for the key

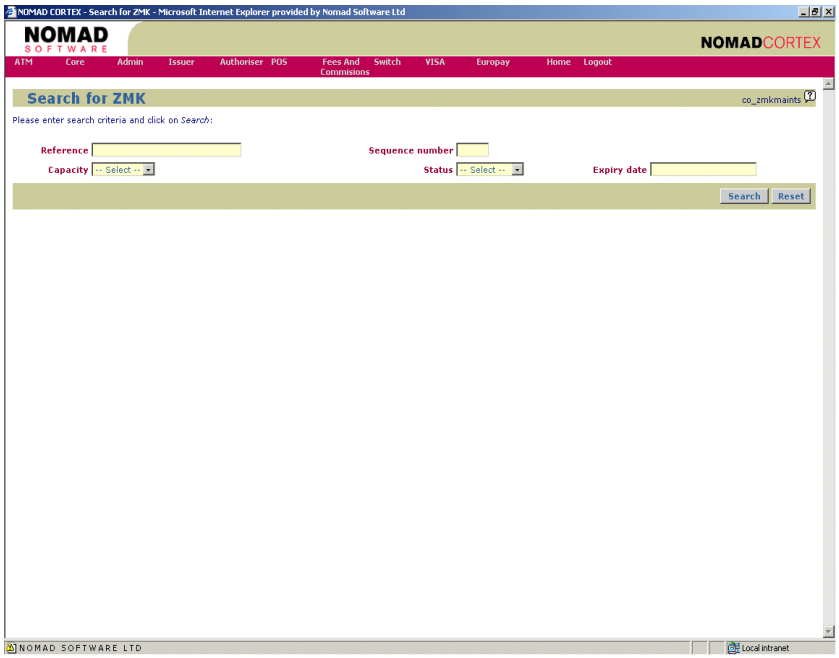
3. At the HSM console enter `FK`.
4. When prompted, enter the following:

Field	Description
Key length	Enter `2`
Key type	Enter `100`
Key scheme	Enter `U`
Component type	Enter `X`
Number of components	Enter `1`
Component 1	Clear component that you noted in step 2
Encrypted key	Encrypted key generated by the HSM. Make a careful note of this.
Key check value	Check value for the key

5. Now repeat steps 1-4 twice more.
6. Now enter all three encrypted components into NOMAD CORTEX by following the procedure described in *To add a ZMK to CORTEX* on page 62.

To view or modify ZMKs

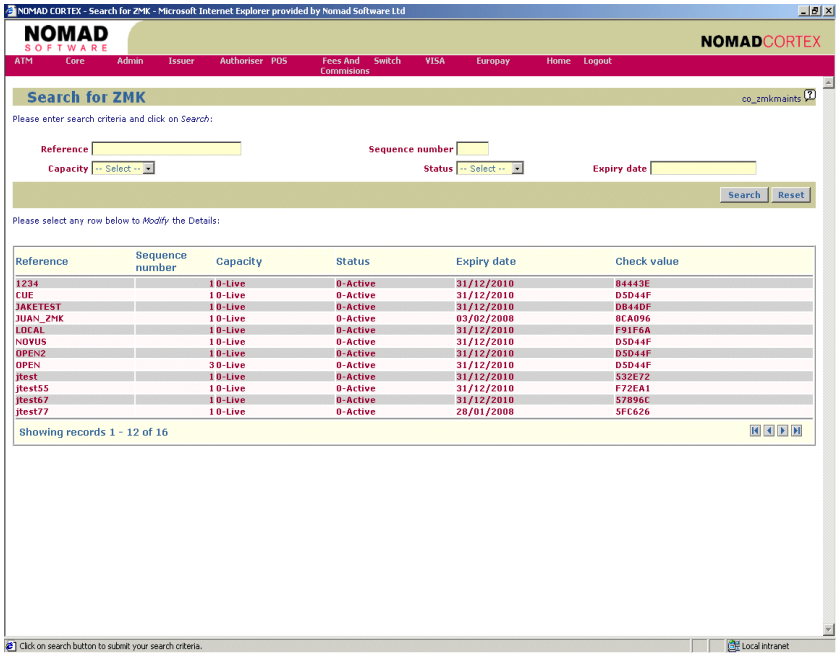
1. From the menu bar, select **Core>Key Maintenance>ZMK Maintenance>Search**. The **Search for ZMK** screen appears:



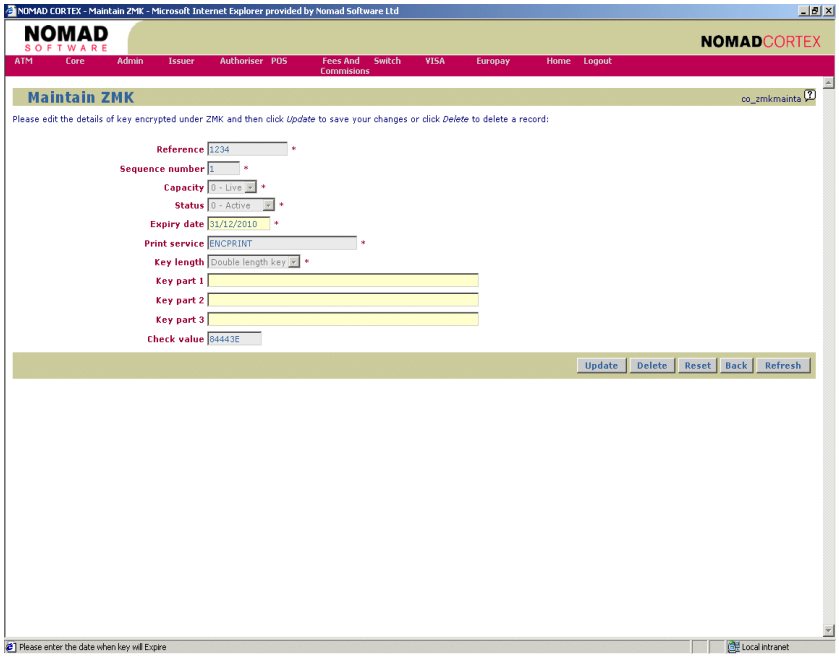
2. In this screen, you can search for a specific key or group of keys using the following fields:

Field	Description	M	O
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✓	✓
Sequence number	Sequence number used to distinguish between ZMKs with the same reference	✓	✓
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✓
Status	Status for the key	✓	✓
Expiry date	Default expiry date of a ZMK	✓	✓

Enter search criteria and click **Search** to view a list of keys matching the criteria, as in the example below:



3. Click on the key that you want to view or modify to display the **Maintain ZMK** screen for that key:



This screen contains the following fields:

Field	Description	M	O
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✗	✗
Sequence number	Sequence number used to distinguish between ZMKs with the same reference	✗	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✗	✗
Status	Status for the key	✗	✗
Expiry date	Default expiry date of the ZMK	✓	✗
Print service	Print service to which new ZMK will be sent	✗	✗
Key length	Whether this is a single or double-length ZMK	✗	✗
Key part 1, Key part 2, Key part 3	For a Visa ZMK (ZMCK), this is the three *encrypted* parts of the received key. Otherwise, leave these fields blank, and NOMAD CORTEX will automatically generate the values.	✓	✓
Check value	Check value of the ZMK	✗	✗

4. When you have finished viewing or modifying, do one of the following:

   ◊ Click **Update** to apply any changes. If you left the Key part 1/2/3 fields blank, NOMAD CORTEX automatically generates the key part values. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,

   ◊ Click **Delete** to delete the key and return to the previous screen, *OR*,

   ◊ Click **Reset** to return any modified fields to their original values, *OR*,

   ◊ Click **Back** to return to the previous screen without saving any changes, *OR*,

   ◊ Click **Refresh** to update the fields of the screen with changes made elsewhere in NOMAD CORTEX

3.4. Maintaining ZPKs

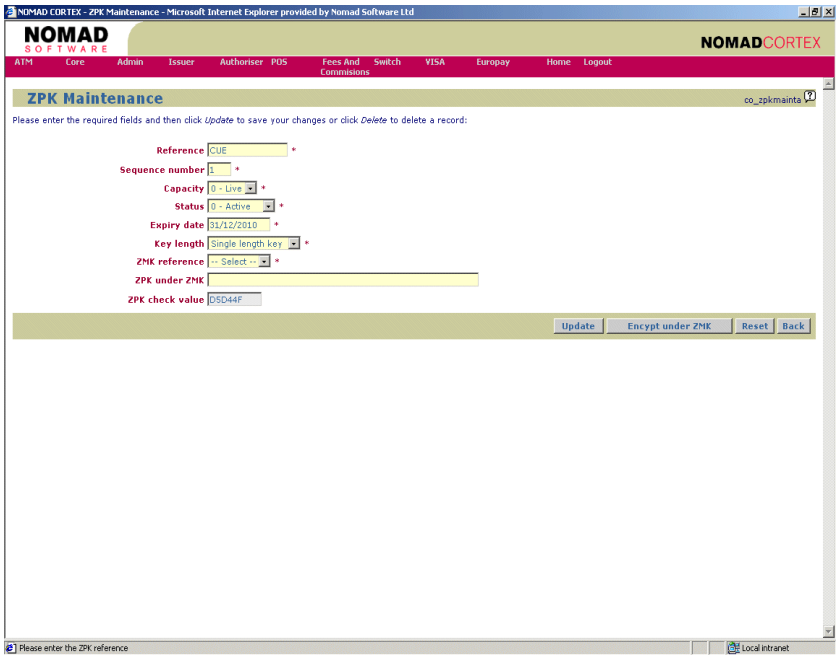
For a full description of the function of ZPKs, see page 20.

You may have to generate a ZPK encrypted under the ZMK, when, for example, you want to send the ZPK to a third party.

A ZPK from another party should be received encrypted under a ZMK that has already been imported into NOMAD CORTEX. The key should also include a check value that can be used to verify the key has been entered correctly.

To add a ZPK to NOMAD CORTEX

1. From the menu bar select **Core>Key Maintenance>ZPK Maintenance>Add**. The **ZPK Maintenance** screen appears:



2. Complete the fields as follows:

Field	Description	M	O
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type. If you are adding a Visa AWK/IWK, type: • `VISAACQ` for an AWK • `VISAISS` for an IWK	✓	✗

Field	Description	M	O
Sequence number	Sequence number used to distinguish between ZPKs with the same reference	✓	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✗
Status	Status for the key	✓	✗
Expiry date	Default expiry date of the ZPK	✓	✗
Key length	Whether this is a single or double-length ZPK	✓	✗
ZMK reference	Reference of the ZMK	✓	✗
ZPK under ZMK	If you are importing a ZPK received from a third party, enter the supplied encryption value of the ZPK under the ZMK. If you are generating a ZPK, leave this field blank, and NOMAD CORTEX will automatically generate the value.	✓	✓
ZPK check value	Check value of the ZPK	✗	✗

3. When you have finished, do one of the following:

◊ Click **Add** to add the ZPK. If the addition is successful, the check value appears in the ZPK check value field and a confirmation message appears:

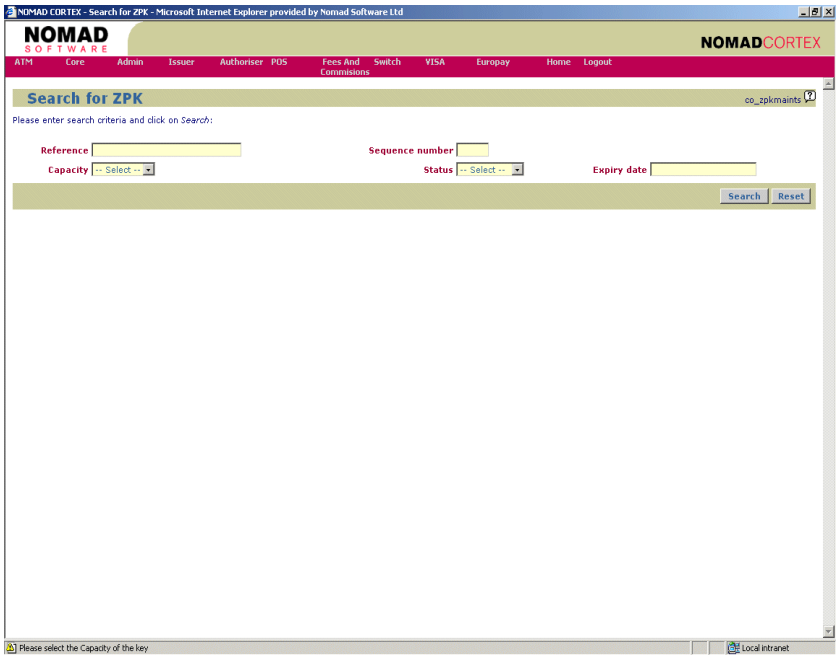
- If you are importing a ZPK received from a third party, verify that ZPK check value is the same as the check value received from them. If they are not identical, then it is likely that you have entered the key incorrectly or entered an incorrect ZMK reference.
- If you are generating a ZPK for export, NOMAD CORTEX automatically generates a value in the ZPK under ZMK field. Carefully note this value, as well as the ZPK check value for sending to the other party.

OR,

◊ Click **Reset** to return any modified fields to their original values

To view or modify ZPKs

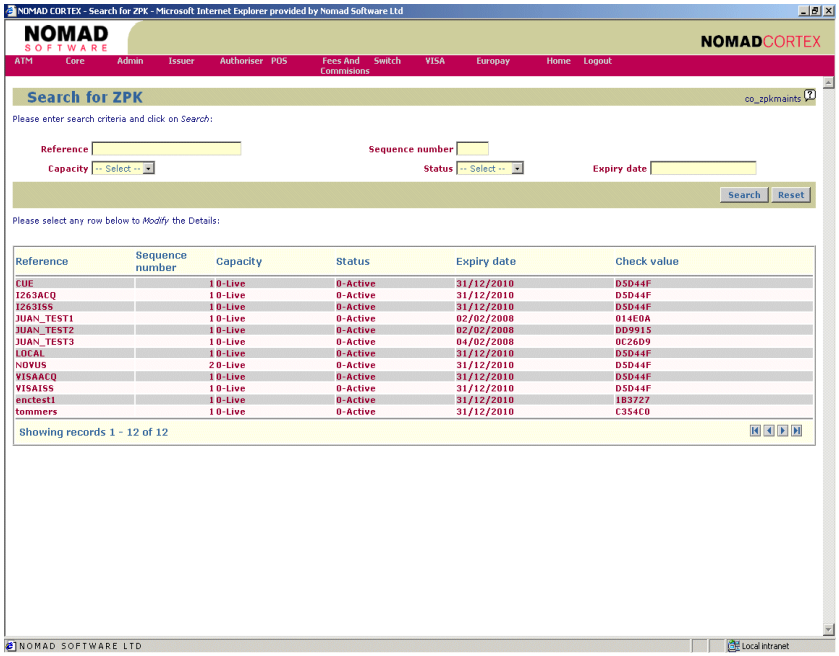
1. From the menu bar, select **Core>Key Maintenance>ZPK Maintenance>Search**. The **Search for ZPK** screen appears:



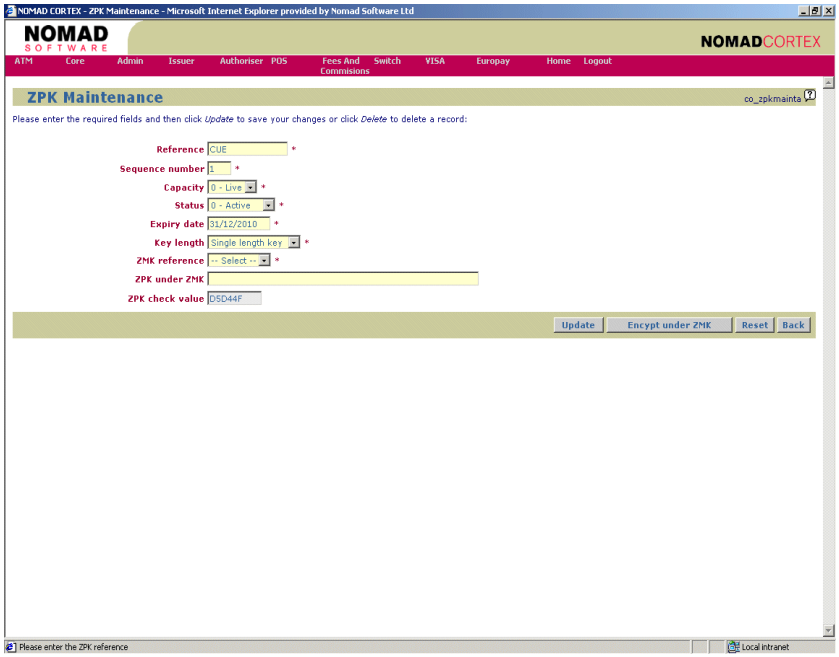
2. In this screen, you can search for a specific key or group of keys using the following fields:

Field	Description	M	O
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✓	✓
Sequence number	Sequence number used to distinguish between ZPKs with the same reference	✓	✓
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✓
Status	Status for the key	✓	✓
Expiry date	Default expiry date of a ZPK	✓	✓

Enter search criteria and click **Search** to view a list of keys matching the criteria, as in the example below:



3. Click on the key that you want to view or modify to display the **ZPK Maintenance** screen for that key:



This screen contains the following fields:

Field	Description	M	O
Reference	NOMAD CORTEX encryption key reference number. Each key must have a reference number. The meaning of the reference number is dependent on key type.	✓	×
Sequence number	Sequence number used to distinguish between ZPKs with the same reference	✓	×
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	×
Status	Status for the key	✓	×
Expiry date	Default expiry date of the ZPK	✓	×
Key length	Whether this is a single or double-length ZPK	✓	×
ZMK reference	Reference of the ZMK	✓	×
ZPK under ZMK	Encryption value of the ZPK under the ZMK. Leave this field blank, and NOMAD CORTEX will automatically generate the value.	✓	✓
ZPK check value	Check value of the ZPK	×	×

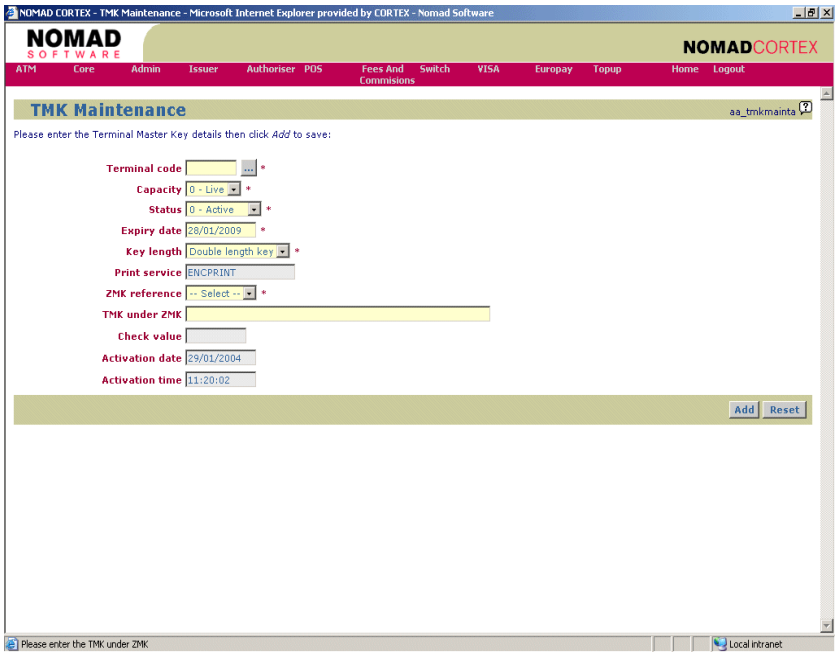
4. When you have finished viewing or modifying, do one of the following:
   - ◊ Click **Add** to add the ZPK. If the addition is successful, the ZPK check value appears in the ZPK check value field and a confirmation message appears, *OR*,
   - ◊ Click **Encrypt under ZMK** to encrypt the key under the Zone Master Key you specified in the ZMK reference field. The Encryption value of the ZPK under the ZMK is displayed in the ZPK under ZMK field. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes

3.5. Maintaining TMKs

For a full description of the function of TMKs, see page 20.

To generate a TMK

1. From the menu bar, select **POS>Maintenance>POS TMK Maintenance>Add** (for POS TMKs), or **ATM>Maintenance>TMK Maintenance>Add** (for ATM TMKs). If you selected ATM TMKs, the **TMK Maintenance** screen appears:



If you selected POS TMKs, this and all subsequent screens basically differ only in title for both key types. Those for ATM TMKs will be used in the examples.

2. Complete the fields as follows:

Field	Description	M	O
Terminal code	Unique identifier for an ATM. Click the Browse button to open a search screen where you can search for an ATM according to the Institution Code, Branch Code and Terminal Number.	✓	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✗
Status	Status for the TMK	✓	✗
Expiry date	Default expiry date of the TMK	✓	✗
Key length	Whether this is a single or double-length TMK. Set to `Double length`.	✓	✗
Print service	Print service to which new TMKs will be sent. Typically, this is `ENCPRINT`.	✗	✗
ZMK reference	Reference of the ZMK	✓	✗
TMK under ZMK	TMK value as encrypted under the ZMK. Leave blank.	✓	✓
Check value	Check value of the TMK	✗	✗
Activation date	Date the TMK was activated	✗	✗
Activation time	Time the TMK was activated	✗	✗

3. Click **Add** to generate the TMK encrypted under the ZMK and add it to the database. If the addition is successful, the TMK appears in the TMK under ZMK field, along with a value in the Check value field, and a confirmation message. Make a note of the Check value. NOMAD CORTEX will print the key at the HSM printer in two parts, for example:

```
Key 1: 0123 4567 89AB CDEF FDEE BC56 4535 33EC
Key 2: CE33 5354 65CB EEDF FEDC BA98 7654 3210
```

You now need to enter the TMK at the terminal (see *To enter a TMK at* an ATM on page 76.

To specify how the TMK is to be printed

Typically, banks want the TMK to be printed in two parts. This is a security measure that means that two trusted officers of the bank would have to collude in order to obtain the full TMK for a terminal.

To specify two-part TMK printing, add the parameter TMK2PART to the ENCRYPT subsection of the NOMAD CORTEX-specific Configuration file ctxcfg. To return to single TMK printing, remove the parameter.

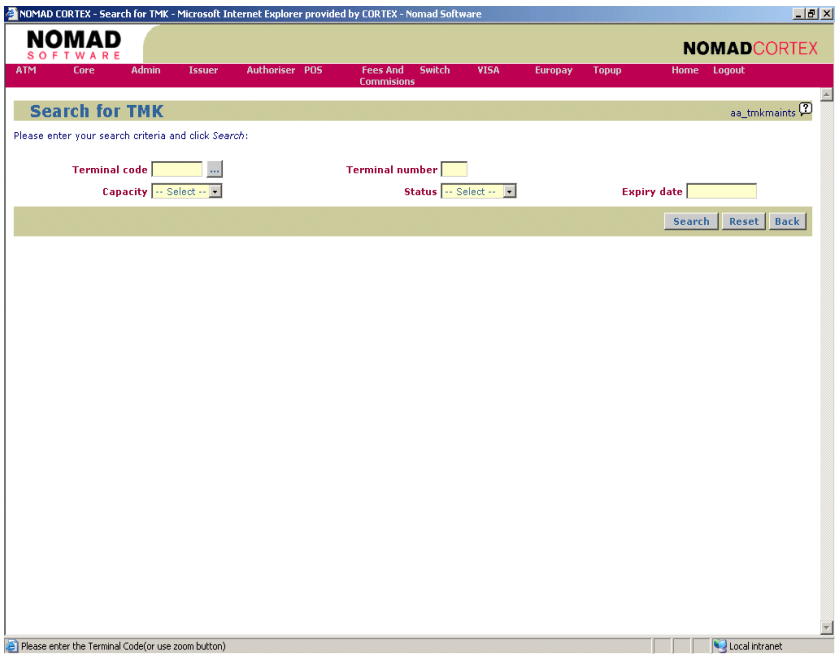
To enter a TMK at an ATM

The precise method used to enter a TMK at a terminal varies according to terminal manufacturer and type. Refer to the relevant manufacturer's documentation for details.

1. Two officers attend the site with the two printed parts of the TMK, plus the noted check value
2. The first officer enters Key 1
3. The second officer enters Key 2
4. The terminal 'X-OR's the two keys internally and then returns a check value. To ensure that the TMK has been entered correctly at the terminal, compare this with the check value returned when the TMK was generated (see *To generate a TMK*). If the check values do not match, transactions will not be communicated correctly.

To view or modify TMK statuses

1. From the menu bar, select **POS>Maintenance>POS TMK Maintenance>Search** (for POS TMKs), or **ATM>Maintenance>TMK Maintenance>Search** (for ATM TMKs). If you selected ATM TMKs, the **Search for TMK** screen appears:

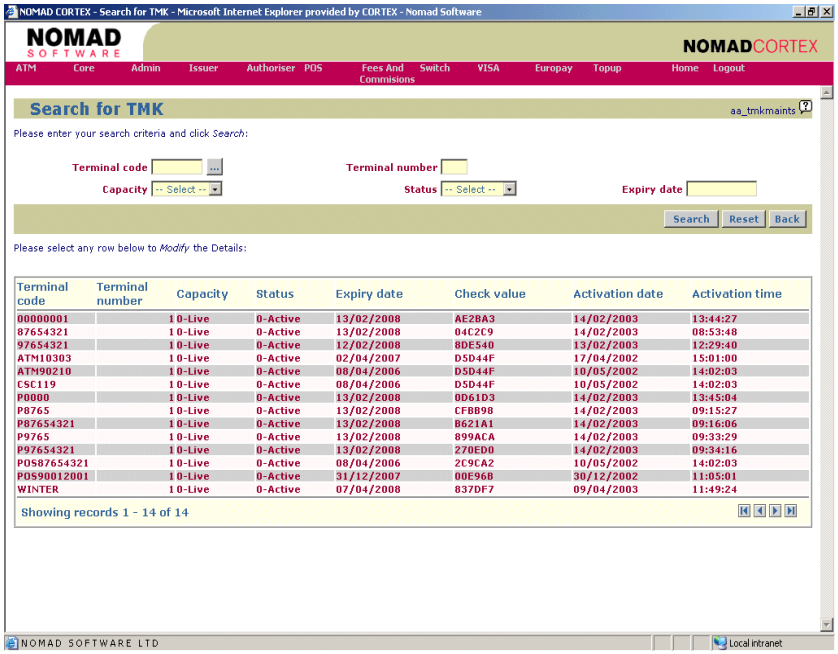


If you selected POS TMKs, this and all subsequent screens basically differ only in title for both key types. Those for ATM TMKs will be used in the examples.

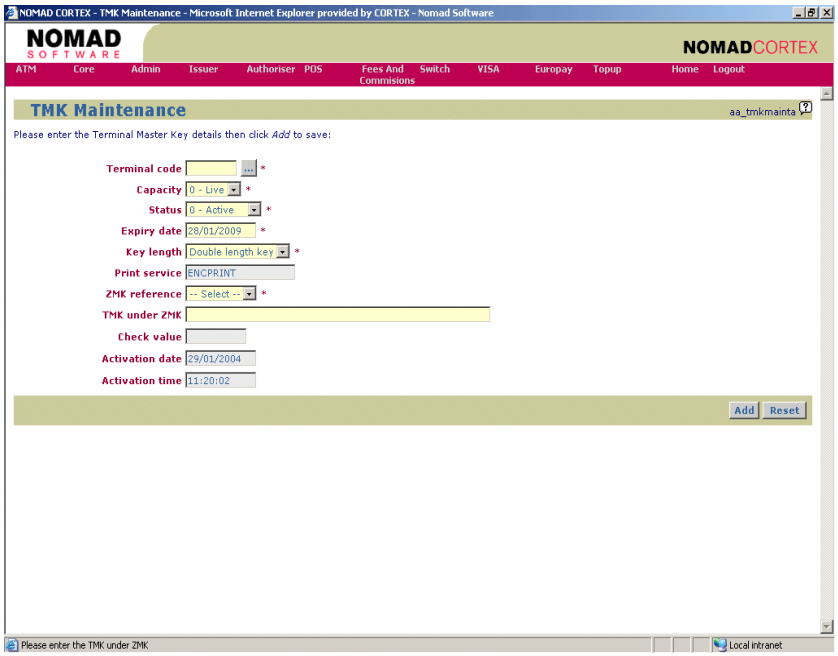
2. In this screen, you can search for a specific key or group of keys using the following fields:

Field	Description	M	O
Terminal code	Unique identifier for an ATM. Click the Browse button to open a search screen where you can search for ATMs.	✓	✓
Terminal number	Sequence number used to distinguish between ATMs in the same branch	✓	✓
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✓
Status	Status for the key	✓	✓
Expiry date	Default expiry date of a TMK	✓	✓

Enter search criteria and click **Search** to view a list of keys matching the criteria, as in the example below:



3. Click on the key that you want to view or modify to display the **TMK Maintenance** screen for that key:



This screen contains the following fields:

Field	Description	M	O
Terminal code	Unique identifier for the ATM. Click the Browse button to open a search screen where you can search for an ATM according to the Institution Code, Branch Code and Terminal Number.	✗	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✗
Status	Status of the TMK	✓	✗
Expiry date	Default expiry date of the TMK	✓	✗
Key length	Whether this is a single or double-length TMK. Set to `Double length`.	✓	✗
Print service	Print service to which new TMKs will be sent	✗	✗
ZMK reference	Reference of the Zone Master Key	✓	✗
TMK under ZMK	TMK value as encrypted under the ZMK	✓	✓
Check value	Check value of the TMK	✗	✗
Activation date	Date the TMK was activated	✗	✗
Activation time	Time the TMK was activated	✗	✗

4. When you have finished viewing or modifying, do one of the following:
   - ◊ Click **Update** to apply any changes. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Delete** to delete the key and return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes, *OR*,
   - ◊ Click **Refresh** to update the fields in the screen with changes made elsewhere in NOMAD CORTEX

To replace TMKs

Before replacing TMKs, you must generate a 'future' ZMK. This ZMK must have the same key value and check value as the active ZMK.

To create a future ZMK, do the following:

1. Open the **Encrypt keys under LMK** screen, as described in *To view or modify encryption keys encrypted under the LMK* in the separate *NOMAD CORTEX Core & Administration User Guide*.
2. Make a note of the values in the Key value and Check value fields of this screen.
3. Open the **Encrypt keys under LMK** screen as described in *To add a previously generated key encrypted under the LMK* in the same document. In the Key value and Check value fields, enter the values you noted before. From the status field, select `Future`. Then add the ZMK.

Now, to replace the TMK, do the following:

1. Generate a new TMK for the ATM as described in *To generate a TMK* on page 74.

   From the Status field of the **TMK Maintenance** screen, select `Future`.

   From the ZMK Reference field, select `Local`.
2. Take the two parts of the printed future TMK to the ATM and enter the keys at the terminal as described in *To enter a TMK at* an ATM on page 76.
3. Request (typically by telephone) an appropriately authorised User at the bank to carry out steps 4-7 below.
4. Open the **ATM Control and Status** screen for the ATM for which you want to replace the TMK, as described in steps 1-3 of *To view the status of an ATM and its associated hardware* in the separate *NOMAD CORTEX ATM User Guide*. Click **Close ATM** to take the ATM out of service.
5. Open the **TMK Maintenance** screen for the future TMK by following the procedure described in *To view or modify TMK statuses* on page 77.
6. Click **Activate future key**. NOMAD CORTEX requests confirmation for the action. If the action is confirmed:
   - ◊ all TMKs for this ATM currently with a status of `Disabled` are deleted
   - ◊ the currently active TMK for this ATM has its status changed to `Disabled`
   - ◊ the future TMK for this ATM has its status changed to `Active`
7. Re-open the **ATM Control and Status** screen for the ATM. Click **Open ATM** to bring the ATM back into service with the new TMK.

To revert to a previously disabled TMK

1. Take the two parts of the printed disabled TMK to the ATM and enter the keys at the terminal as described in *To enter a TMK at* an ATM on page 76.
2. Request (typically by telephone) an appropriately authorised User at the bank to carry out steps 3-6 below.
3. Open the **ATM Control and Status** screen for the ATM for which you want to revert to the previously disabled TMK, as described in steps 1-3 of *To view the status of an ATM and its associated hardware* in the separate *NOMAD CORTEX ATM User Guide*. Click **Close ATM** to take the ATM out of service.
4. Open the **TMK Maintenance** screen for the disabled TMK by following the procedure described in *To view or modify TMK statuses* on page 77.
5. Click **Activate disabled key**. NOMAD CORTEX requests confirmation for the action. If the action is confirmed:
   - ◊ the currently active TMK for this ATM has its status changed to `Disabled`
   - ◊ the previously disabled TMK for this ATM has its status changed to `Active`
6. Re-open the **ATM Control and Status** screen for the ATM. Click **Open ATM** to bring the ATM back into service with the previous TMK.

3.6. Generating TPKs

For a full description of the function of TPK, see page 21.

Note: Although TPKs are sent automatically, you could cause a new key to be sent by simply switching an ATM into supervisor mode, then exiting this mode.

3.7. Maintaining CVKs and PVKs

For a full description of the function of CVKs and PVKs, see pages 21-22.

NOMAD CORTEX can support up to ten PVKs per BIN. This allows cards issued under the same BIN to use different PVKs to generate their PVVs, thus providing an additional level of security.

The PVK Index (PVKI) is written onto Track 2 of the card. The PVKI that NOMAD CORTEX currently uses for PVV generation is defined in the card format of the BIN (see *To view or modify card formats* in the separate document *NOMAD CORTEX Issuer User Guide*).

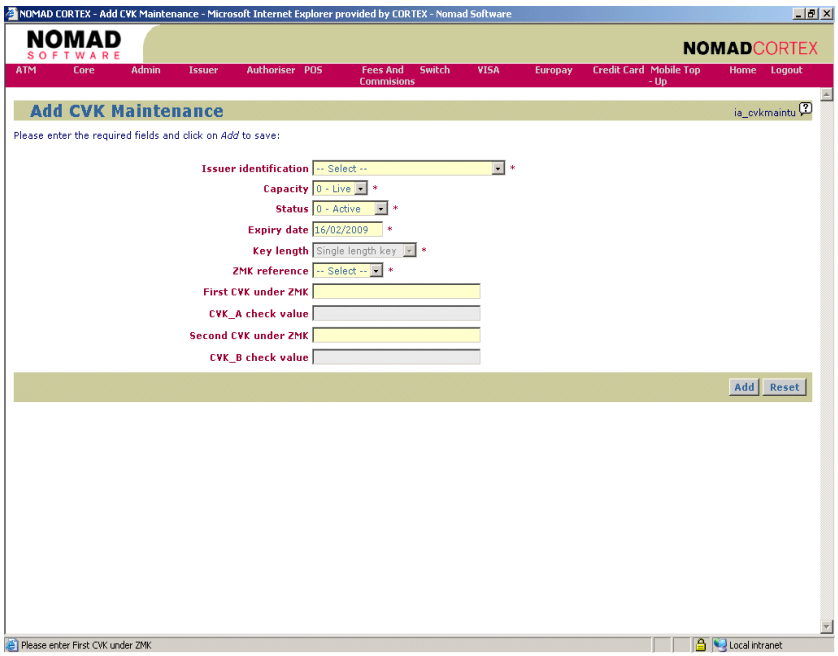
To add a CVK or PVK pair to NOMAD CORTEX

1. According to the type of key pair that you want to add, do one of the following:

   ◊ For CVKs, from the menu bar, select **Issuer>Key Maintenance>Card Verification Key>Add**.

   ◊ For PVKs, from the menu bar, select **Issuer>Key Maintenance>PIN Verification Key>Add**.

   If you selected Card Verification Key, the **Add CVK Maintenance** screen appears:



   If you selected PIN Verification Key, this and all subsequent screens differ only in title from those for CVKs given in the following examples.

2. Complete the fields as follows:

Field	Description	M	O
Issuer identification	BIN of cards for which the CVK is to be created	✓	×
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	×
Status	Status for the key	✓	×
Expiry date	Expiry date of the CVK	✓	×
Key length	Whether this is a single-length or double-length CVK	×	×
ZMK reference	Reference of the ZMK	✓	×
First CVK under ZMK	First encryption value of CVK pair. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
CVK_A check value	Check value of first CVK	×	×
Second CVK under ZMK	Second encryption value of CVK pair. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
CVK_B check value	Check value of second CVK	×	×

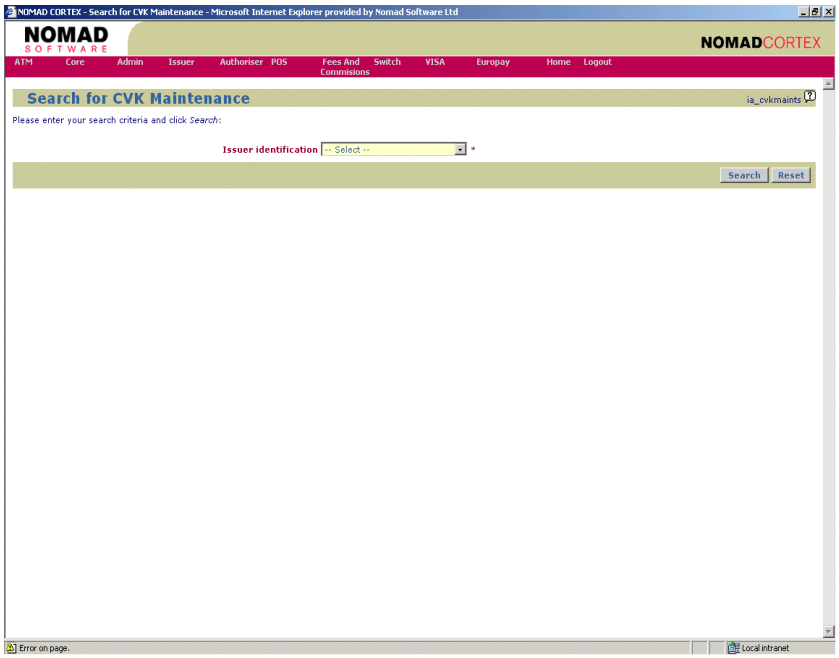
3. When you have finished, do one of the following:

   ◊ Click **Add** to add the CVK. If you left the First CVK under ZMK/Second CVK under ZMK fields blank, NOMAD CORTEX automatically generates the values. If the addition is successful, the CVK check values are completed and a confirmation message appears, *OR*,

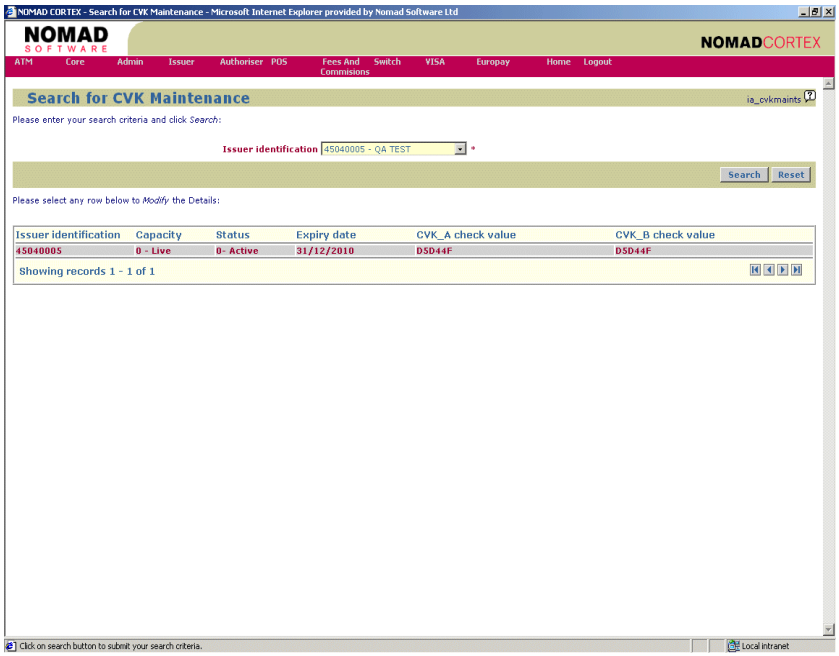
   ◊ Click **Reset** to return any modified fields to their original values

To view or modify CVKs

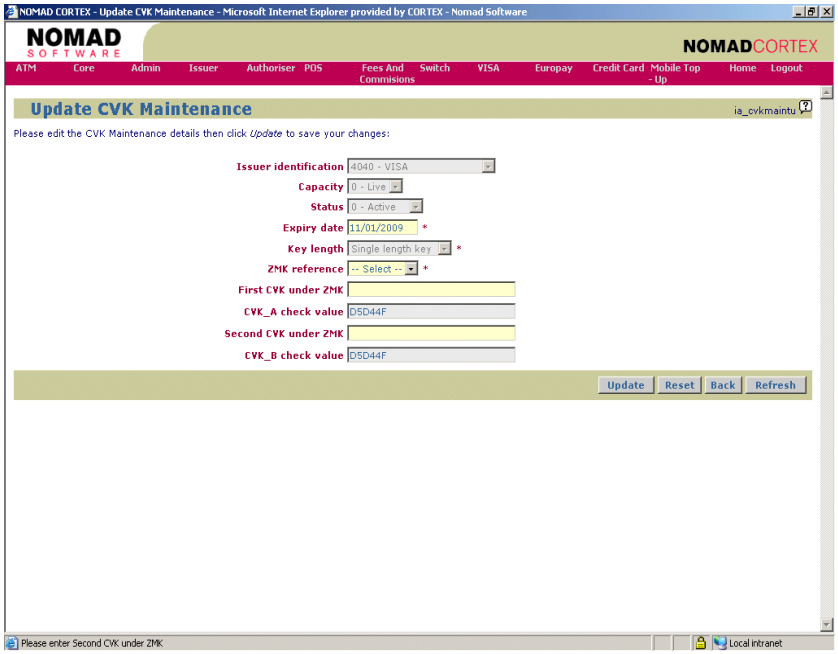
1. From the menu bar, select **Issuer>Key Maintenance>Card Verification Key>Search**. The **Search for CVK Maintenance** screen appears:



2. From the Issuer Identification field, select the bank's unique issuer identification and then click **Search** to display a list of all CVKs for the issuer, as in the example below:



3. Click on the key that you want to view or modify to display the **Update CVK Maintenance** screen for that key:



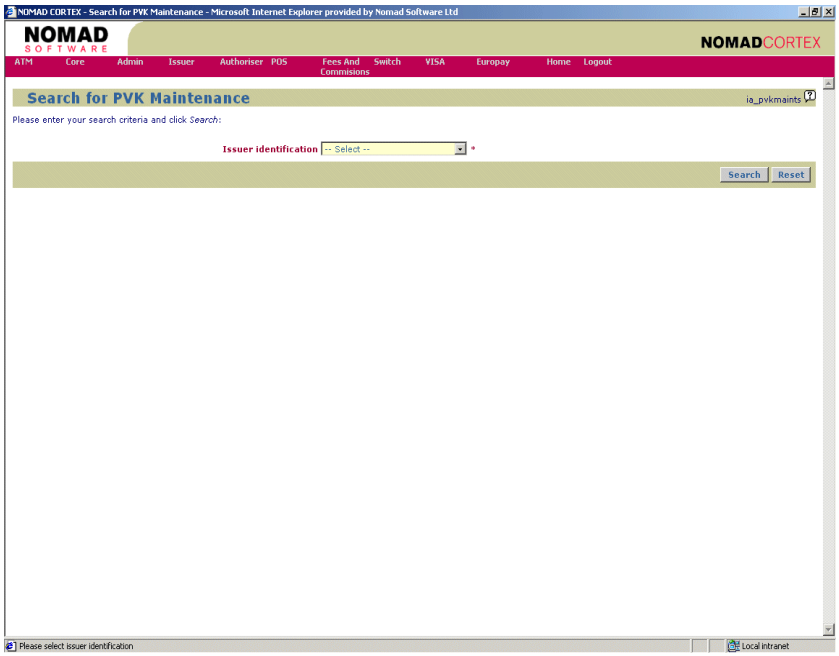
This screen contains the following fields:

Field	Description	M	O
Issuer identification	Identifier for the institution issuing cards	✗	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✗	✗
Status	Status for the key	✗	✗
Expiry date	Expiry date of the CVK	✓	✗
Key length	Whether this is a single-length or double-length CVK	✗	✗
ZMK reference	Reference of the ZMK	✓	✗
First CVK under ZMK	First encryption value of a pair. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
CVK_A check value	Check value of first CVK	✗	✗
Second CVK under ZMK	Second encryption value of a pair. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
CVK_B check value	Check value of second CVK	✗	✗

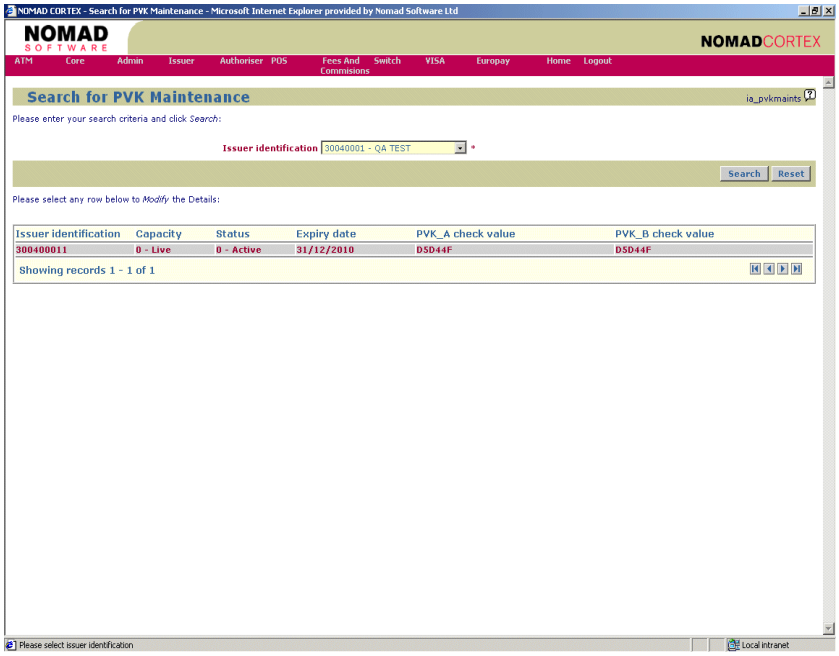
4. When you have finished viewing or modifying, do one of the following:
   - ◊ Click **Update** to apply any changes. If you left the First CVK under ZMK/Second CVK under ZMK fields blank, NOMAD CORTEX automatically generates the values. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes, *OR,*
   - ◊ Click **Refresh** to update the fields in the screen with changes made elsewhere in NOMAD CORTEX

To view or modify PVKs

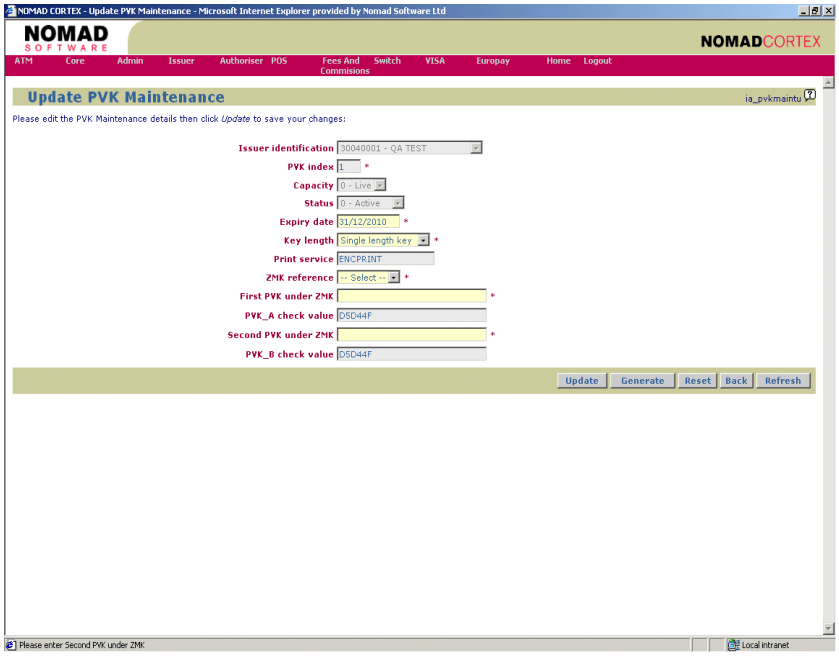
1. From the menu bar, select **Issuer>Key Maintenance>PIN Verification Key>Search**. The **Search for PVK Maintenance** screen appears:



2. From the Issuer Identification field, select the bank's unique issuer identification and then click **Search** to display a list of all PVKs for the issuer, as in the example below:



3. Click on the key that you want to view or modify to display the **Update PVK Maintenance** screen for that key:



Complete the fields as follows:

Field	Description	M	O
Issuer identification	Identifier for the institution issuing cards	✗	✗
PVK index	PVK index (for Visa cards)	✗	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✗	✗
Status	Status for the key	✗	✗
Expiry date	Expiry date of the PVK	✓	✗
Key length	Whether this is a single-length or double-length PVK	✗	✗
Print service	Print service to which new PVKs will be sent	✗	✗
ZMK reference	Reference of the ZMK	✓	✗
First PVK under ZMK	First encryption value of a pair. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
PVK_A check value	Check value of first PVK	✗	✗

Field	Description	M	O
Second PVK under ZMK	Second encryption value of a pair. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
PVK_B check value	Check value of second PVK	✗	✗

4. When you have finished viewing or modifying, do one of the following:
   - ◊ Click **Update** to apply any changes. If you left the First PVK under ZMK/Second PVK under ZMK fields blank, NOMAD CORTEX automatically generates the values. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes, *OR,*
   - ◊ Click **Refresh** to update the fields in the screen with changes made elsewhere in NOMAD CORTEX

To send a PVK/CVK to a third party

You send PVK/CVK pairs to third parties (such as Visa) as encrypted values under the ZMK only. These would have been provided by following the procedure described in *To add a CVK or PVK pair to NOMAD CORTEX* on page 82.

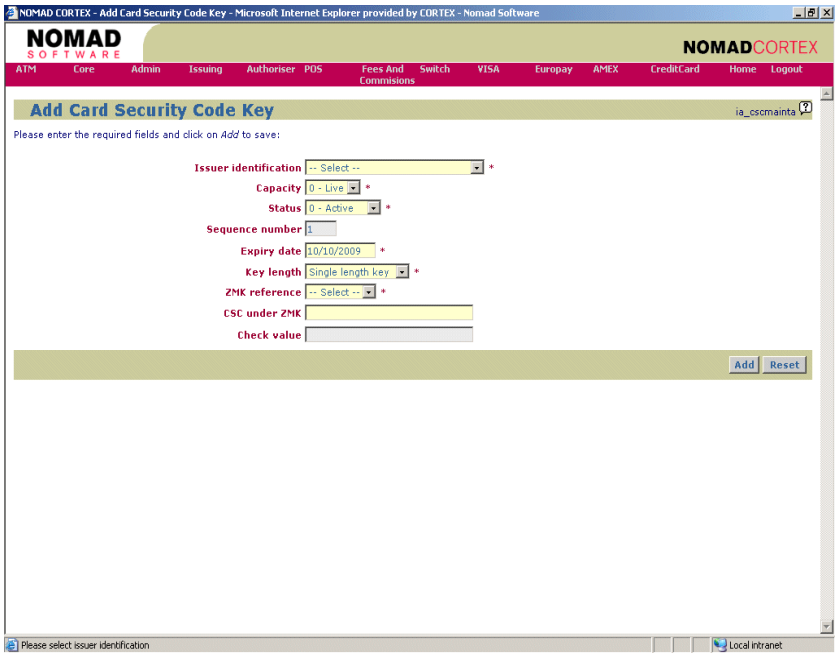
Typically, you fax the two parts of a PVK/CVK pair to the third party. Refer to the appropriate security authority at your bank for the correct procedure to follow.

3.8. Maintaining CSCKs

For a full description of the function of CSCKs, see page 22.

To add a CSCK to NOMAD CORTEX

1. From the menu bar, select **Issuer>Key Maintenance>Card Security Code Key>Add**. The **Add Card Security Code Key** screen appears:



2. Complete the fields as follows:

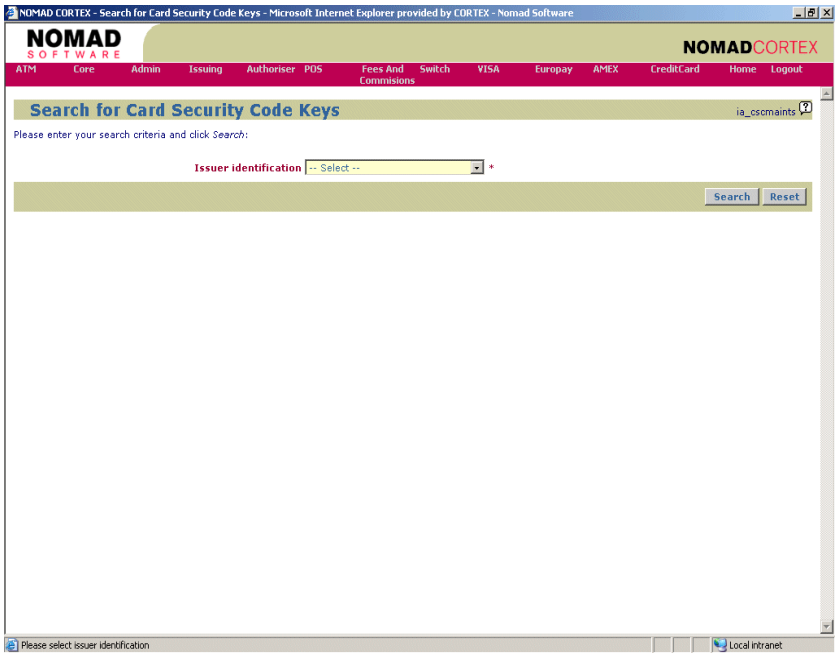
Field	Description	M	O
Issuer identification	Identifier for the institution issuing cards	✓	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	✗
Status	Status of the CSCK	✓	✗
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	✗	✗
Expiry date	Expiry date of the CSCK	✓	✗
Key length	Whether this is a single-length or double-length CSCK	✓	✗

Field	Description	M	O
ZMK reference	Reference of the CSCK	✓	✗
CSC under ZMK	Encryption value of the CSCK under the ZMK. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
Check value	Check value of the CSCK	✗	✗

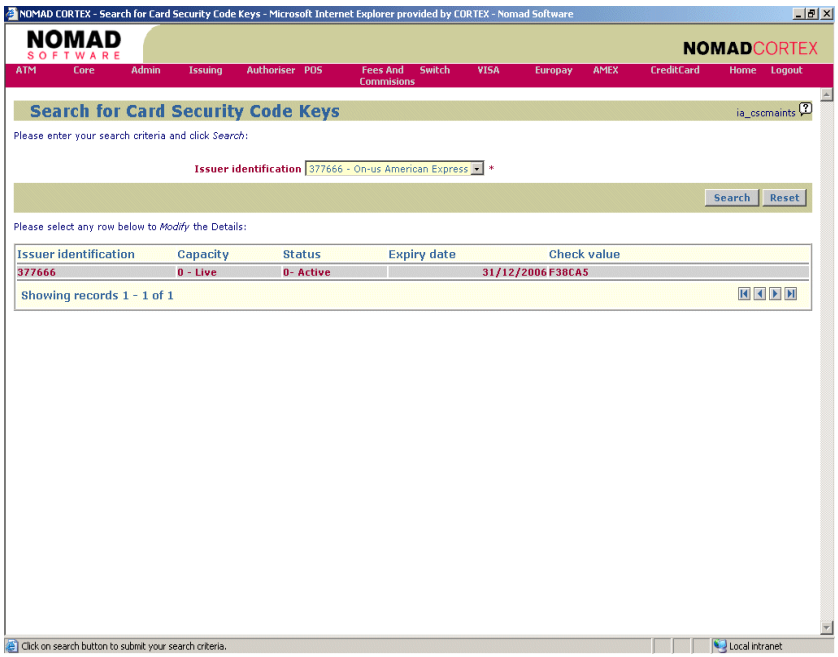
3. When you have finished, do one of the following:
   - ◊ Click **Add** to add the CSCK. If you left CSC under ZMK field blank, NOMAD CORTEX automatically generates the values. If the addition is successful, the Card Security Code Key check value is completed and a confirmation message appears, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values

To view or modify CSCKs

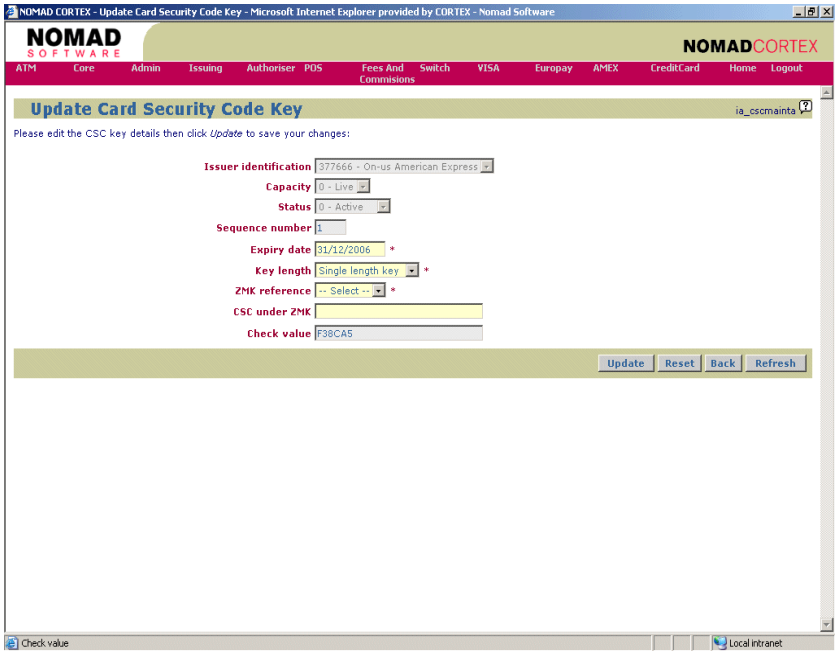
1. From the menu bar, select **Issuer>Key Maintenance>Card Security Code Key>Search**. The **Search for Card Security Code Keys** screen appears:



2. From the Issuer Identification field, select the bank's unique issuer identification and then click **Search** to display a list of all CSCKs for the issuer, as in the example below:



3. Click on the key that you want to view or modify to display the **Update Card Security Code Key** screen for that key:



Complete the fields as follows:

Field	Description	M	O
Issuer identification	Identifier for the institution issuing cards	✗	✗
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✗	✗
Status	Status of the CSCK	✗	✗
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	✗	✗
Expiry date	Expiry date of the CSCK	✓	✗
Key length	Whether this is a single-length or double-length CSCK	✗	✗
ZMK reference	Reference of the CSCK	✓	✗
CSC under ZMK	Encryption value of the CSCK under the ZMK. Leave blank and NOMAD CORTEX will automatically generate the value.	✓	✓
Check value	Check value of the CSCK	✗	✗

4. When you have finished viewing or modifying, do one of the following:
   - ◊ Click **Update** to apply any changes. If you left the CSC under ZMK field blank, NOMAD CORTEX automatically generates the values. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values, *OR*,
   - ◊ Click **Back** to return to the previous screen without saving any changes, *OR,*
   - ◊ Click **Refresh** to update the fields in the screen with changes made elsewhere in NOMAD CORTEX

To send a CSCK to a third party

You send CSCKs to third parties as encrypted values under the ZMK only. These would have been provided by following the procedure described in *To add a CSCK to NOMAD CORTEX* on page 90.

Refer to the appropriate security authority at your bank for the correct procedure to follow.

3.9. Maintaining IMKs

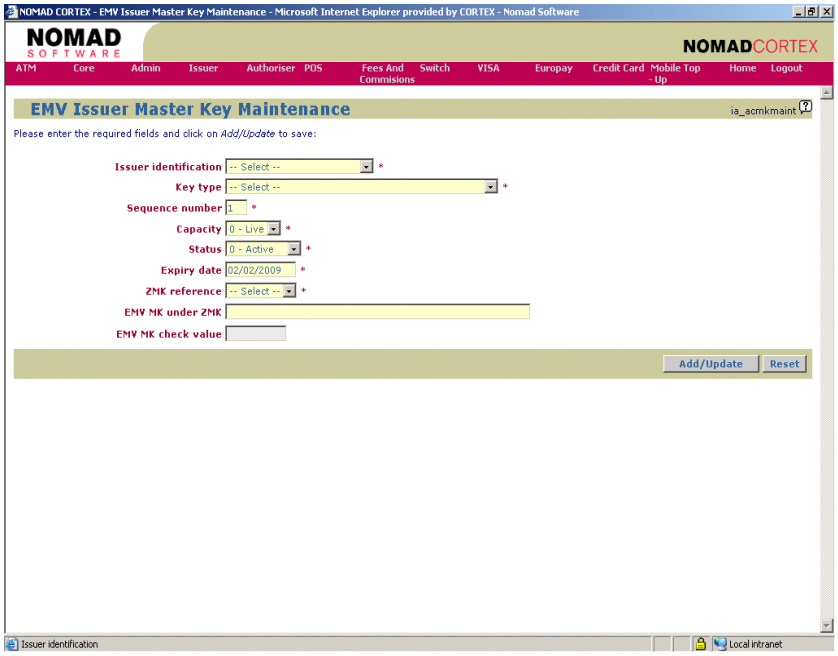
For a full description of the function of IMKs, see page 22.

To add a previously generated IMK to NOMAD CORTEX

The procedure for adding an IMK to NOMAD CORTEX is the same as that described in *To add a previously generated key encrypted under the LMK* on page 54.

To import an IMK encrypted under a ZMK

1. From the menu bar, select **Issuer>Key Maintenance>Issuer Cryptogram Maintenance**. The **EMV Issuer Master Key Maintenance** screen appears:



Complete the fields as follows:

Field	Description	M	O
Issuer identification	Identifier for the institution issuing cards	✓	×
Key type	Type of Issuer Master Key: ACMK= Application Cryptogram Master Key SMCMK= Secure Messaging with Confidentiality Master Key SMIMK= Secure Messaging with Integrity Master Key DACMK= Data Authentication Code Master Key DNMK= Dynamic Number Master Key	✓	×
Sequence number	Sequence number used to distinguish between NOMAD CORTEX encryption keys that have the same Key type and Reference	✓	×
Capacity	Whether this is a live or test key. Test keys should never be used on a live system and vice-versa.	✓	×
Status	Status for the key	✓	×
Expiry date	Expiry date of the Issuer Master Key	✓	×
ZMK reference	Reference of the Issuer Master Key	✓	×
EMV MK under ZMK	IMK encrypted under the ZMK. Leave blank and NOMAD CORTEX will automatically generate a value.	✓	✓
EMV MK check value	Check value of the EMV MK	×	×

2. When you have finished, do one of the following:

   ◊ Click **Add/Update** to add the key under the ZMK. If you left the EMV MK under ZMK field blank, NOMAD CORTEX automatically generates the value. If the addition is successful, the check value appears in the EMV MK check value field and a confirmation message appears, *OR*,

   ◊ Click **Reset** to return any modified fields to their original values

To export an existing ACMK

To export an existing ACMK under a ZMK for sending to a third party, you need to know the encrypted ACMK as stored on the NOMAD CORTEX database. You also need to know the encrypted ZMK with which you want to re-encrypt the ACMK.

1. Ensure that export of keys under the 'X' scheme is enabled on the HSM. If you are unsure as to whether this is the case, enter `CS` at the HSM console, and check that the field Enable X9.17 for export is set to `Y`.
2. Open the **Encrypt keys under LMK** screen for the ZMK (Key type=1), as described in steps 1-3 of *To view or modify encryption keys encrypted under the LMK* on page 58. Make a note of the encrypted key from the Key value field, and the Check value from the Check value field.
3. Open the **Encrypt keys under LMK** screen for the ACMK (Key type=11). Again, make a note of the encrypted key from the Key value field, and the Check value from the Check value field.
4. Enter `KE` at the HSM console.
5. When prompted, enter the following:

Field	Description
Key type	Enter `109`
Key Scheme (ZMK)	ZMK scheme. This is the type of encryption that the HSM uses for the ZMK. Enter `X` to indicate encryption of a double-length key using ANSI X9.17 methods only available for import and export of keys.
ZMK	Encrypted ZMK. Enter `U` followed by the encrypted ZMK, for example: `U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX`
Key	Encrypted ACMK that you want to export. Enter `U` followed by the encrypted ACMK, for example: `U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY`
Key under ZMK	ACMK encrypted under the ZMK. Enter `X` followed by the encrypted ACMK, for example: `X AAAA AAAA AAAA AAAA AAAA AAAA AAAA AAAA`
Key check value	Check value for the key

6. To ensure that the procedure was successful, verify that the check value of the ACMK is the same as the check value generated by the HSM.

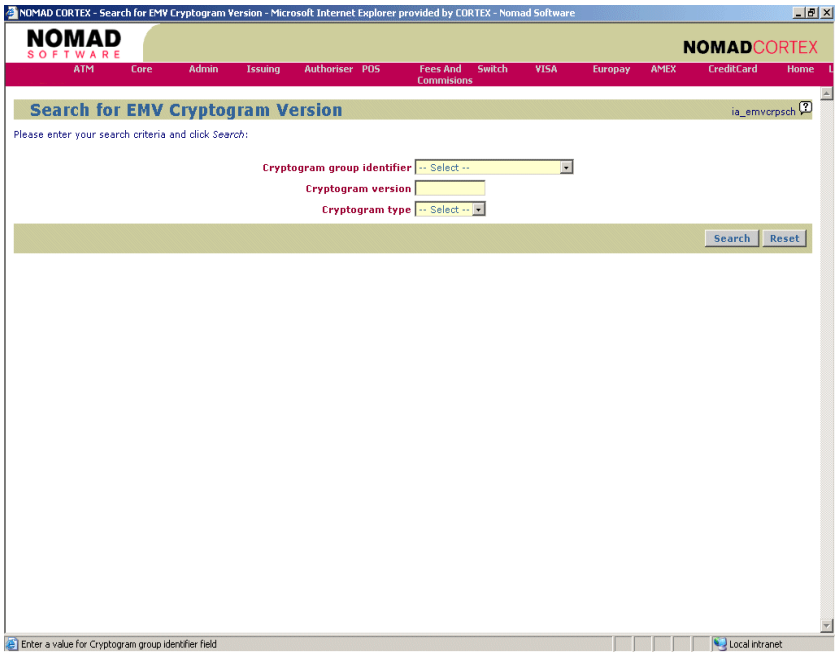
7. Now the ACMK encrypted under the ZMK, and the check value can be sent to the third party. When sending the ACMK, leave out the initial X.

3.10. Maintaining EMV Cryptogram Version Definitions

EMV cryptogram version definitions specify how application cryptograms are built—in other words, which EMV fields (and any other extra fixed data) are used, and in which order.

To view or modify the definition of EMV cryptogram versions

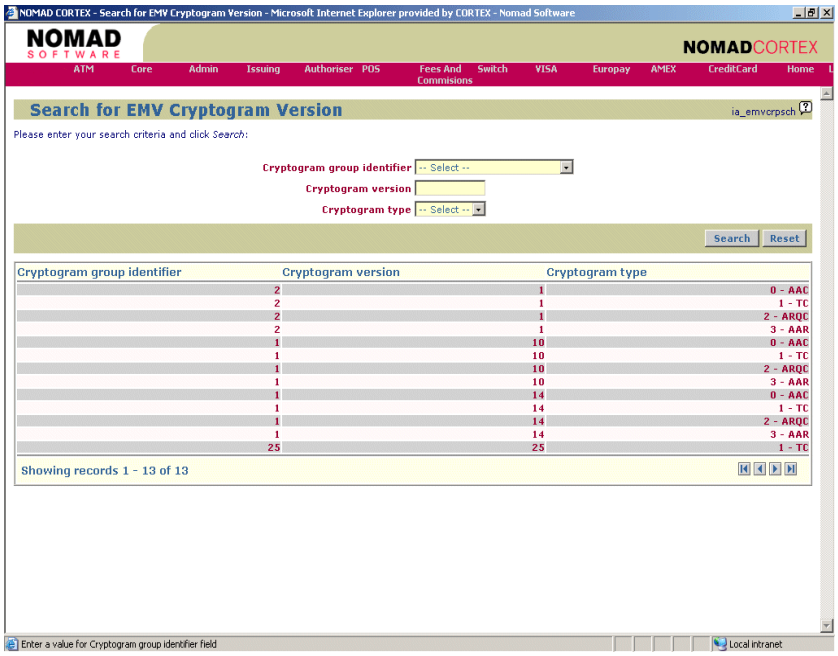
1. From the menu bar, select **Authoriser>EMV>EMV Cryptogram>Search**. The **Search for EMV Cryptogram Version** screen appears:



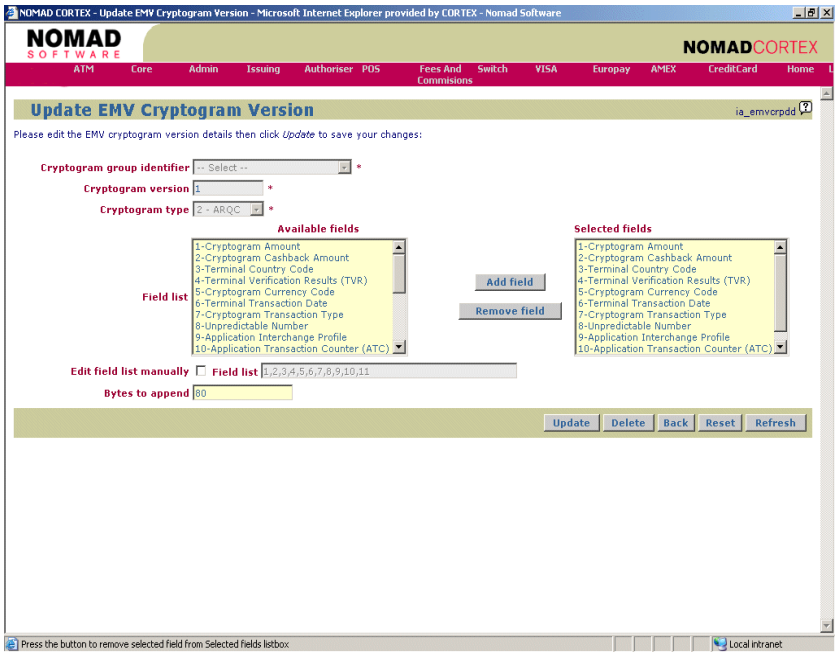
2. In this screen, you can search for a specific cryptogram using the following fields:

Field	Description	M	O
Cryptogram group identifier	Unique identifier for a cryptogram group	✓	✓
Cryptogram version	Cryptogram version	✓	✓
Cryptogram type	Type of cryptogram: 0= AAC 1= TC 2= ARQC 3= AAR	✓	✓

Enter search criteria and click **Search** to view a list of cryptograms matching the criteria, as in the example below:



3. Click on the cryptogram for which you want to view or modify the version definition to display the **Update EMV Cryptogram Version** screen:



This screen contains the following fields:

Field	Description	M	O
Cryptogram group identifier	Unique identifier for the group to which the cryptogram belongs. For example, all Visa cards could use group 1, Plus cards group 2, MasterCards group 3, and so on. The group identifier is used as cryptogram version 10 for one card product may be different to cryptogram version 10 for another.	✗	✗
Cryptogram version	Cryptogram version. When a transaction arrives, the value in C_CRPTVER (in 9F10 Issuer authentication data) indicates which cryptogram version the card uses	✗	✗
Cryptogram type	Type of Cryptogram. This enables, for example, an AAC to be constructed from a different number or order of fields than those for an AAR cryptogram. Typically, records should be set up for each cryptogram type with the field list (below) the same: 0= AAC 1= TC 2= ARQC 3= AAR	✗	✗
Available fields	Fields that are available for this cryptogram version	✓	✓
Selected fields	Fields that are present in this cryptogram version	✓	✓
Field list	Fields that are present in this cryptogram version expressed as a comma-separated list of integers. To synchronise this list with the fields selected under Selected fields, click **Refresh**. If you select Edit field list manually, you can manually edit the fields. You should pay attention to the order of fields, as, if these are incorrect, then verification will not work correctly.	✗	✗

Field	Description	M	O
Edit field list manually	Select this to manually edit the fields in Field list above	✓	✓
Bytes to append	Any fixed bytes to append to the end of the field list data before MAC generation. The format is a string of hexadecimal digits, with the default being an empty string. For example, MasterCards and Visa cryptogram version 14 generally require a single 0x80 byte to be appended. So, you would enter `80` in this case. This field must be correct or verification will not work.	✓	✓

4. Do one of the following:

◊ Modify the definition of the cryptogram version by doing one of the following:

- To add a field for the cryptogram version, select it from the list under Available fields, and then click **Add field**. The field appears in the list under Selected fields.
- To remove a field from the cryptogram version, select it from the list under Selected fields, and then click **Remove field**

◊ Click **Update** to apply any changes. If the update is applied, a confirmation message appears. Then click **Back** to return to the previous screen, *OR*,

◊ Click **Delete** to delete the cryptogram version and return to the previous screen, *OR*,

◊ Click **Reset** to return any modified fields to their original values, *OR*,

◊ Click **Back** to return to the previous screen without saving any changes, *OR*,

◊ Click **Refresh** to update the fields in the screen with changes made elsewhere in NOMAD CORTEX

To add a definition for an EMV cryptogram version

1. From the menu bar, select **Authoriser>EMV>EMV Cryptogram>Add**. The **Add EMV Cryptogram Version** screen appears. This screen contains the same fields as the **Update EMV Cryptogram Version** screen.
2. Complete the fields as follows:

Field	Description	M	O
Cryptogram group identifier	Unique identifier for the group to which the cryptogram belongs. For example, all Visa cards could use group 1, Plus cards group 2, MasterCards group 3, and so on. The group identifier is used as cryptogram version 10 for one card product may be different to cryptogram version 10 for another.	✓	✗
Create new group ID	Select this check box to specify that a new cryptogram group is to be created	✓	✓
Group ID	Unique identifier for the new cryptogram group with which the cryptogram version is to be associated. This field is activated if you select Create new group ID.	✓	✓
Description	Brief textual description for the new cryptogram group with which the cryptogram version is to be associated. This field is activated if you select Create new group ID.	✓	✓
Cryptogram version	Cryptogram version. When a transaction arrives, the value in C_CRPTVER (in 9F10 Issuer authentication data) indicates which cryptogram version the card uses	✓	✗
Cryptogram type	Type of Cryptogram. This enables, for example, an AAC to be constructed from a different number or order of fields than those for an AAR cryptogram. Typically, records should be set up for each cryptogram type with the field list (below) the same: 0= AAC 1= TC 2= ARQC 3= AAR	✓	✗

Field	Description	M	O
Available fields	Fields that are available for this cryptogram version	✓	✓
Selected fields	Fields that are present in this cryptogram version	✓	✓
Field list	Fields that are present in this cryptogram version expressed as a comma-separated list of integers. To synchronise this list with the fields selected under Selected fields, click **Refresh**. If you select Edit field list manually, you can manually edit the fields. You should pay attention to the order of fields, as, if these are incorrect, then verification will not work correctly.	✗	✗
Edit field list manually	Select this to manually edit the fields in Field list above	✓	✓
Bytes to append	Any fixed bytes to append to the end of the field list data before MAC generation. The format is a string of hexadecimal digits, with the default being an empty string. For example, MasterCards and Visa cryptogram version 14 generally require a single 0x80 byte to be appended. So, you would enter `80` in this case. This field must be correct or verification will not work.	✓	✓

3. When you have finished, do one of the following:
   - ◊ Click **Add** to add the cryptogram version definition to the database. If the addition is successful, a confirmation message appears, *OR*,
   - ◊ Click **Reset** to return any modified fields to their original values

3.11. Generating MAC keys (TAKs)

For a full description of the function of LMKs, see page 21.

Send Us Your Comments

NOMAD CORTEX Key Management Operations Guide Version 2.5

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